



Machine instructions and Addressing modes. ALU, data-path and control unit. Instruction pipelining. **Pipeline hazards**, Memory hierarchy: cache, main memory and secondary storage; I/O interface (Interrupt and DMA mode)

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	3	1	2	1	2	2	2	3	1	2	1
2 Marks Count	4	5	3	4	3	3	4	2	2	2	2
Total Marks	11	11	8	9	8	8	10	7	5	6	5

Welcome to the "Computer Organization & Architecture" chapter of your GATE Computer Science preparation. This crucial subject delves into the fundamental design and operational principles of a computer system, from the lowest-level hardware components to how instructions are executed and data is managed. Understanding COA is vital for any computer scientist as it provides the bedrock knowledge for designing efficient systems, writing optimized code, and comprehending the performance implications of software choices. In the GATE CS exam, COA typically carries a weightage of 8-12 marks, with questions ranging from conceptual understanding of architectural principles (like CISC vs. RISC, control unit types) to numerical problems involving cache memory, pipelining performance, virtual memory, and disk access times. Expect a mix of Multiple Choice Questions (MCQs), Multiple Select Questions (MSQs), and Numerical Answer Type (NAT) questions, often requiring precise calculations and a deep grasp of underlying mechanisms.

Topic-wise Key Concepts

Addressing Modes

Addressing modes define how the effective address of an operand is calculated. They specify where the operand is located, whether in a register, memory, or as part of the instruction itself, enabling flexible and efficient memory access.

Important Formulas & Concepts

- **Immediate:** Operand is part of the instruction. $EA = \text{Operand Value}$. No memory access for operand.
- **Direct:** $EA = \text{Address}$ (address field in instruction). One memory access to fetch operand.
- **Indirect:** $EA = M[\text{Address}]$ (address field points to memory location containing EA). Two memory accesses (one for EA, one for operand).
- **Register:** Operand is in a specified register. $EA = R$. No memory access.
- **Register Indirect:** $EA = [R]$ (register contains EA). One memory access to fetch operand.
- **Relative (PC-relative):** $EA = PC + \text{Offset}$ (offset from Program Counter). Used for branch instructions.
- **Indexed:** $EA = \text{Base_Address} + \text{Index_Register}$. Used for array access.
- **Base-Register:** $EA = \text{Base_Register} + \text{Offset}$. Used for relocating programs.

Key Properties & Identities

- Trade-offs exist between instruction length, execution speed, and flexibility.
- Used to access data, implement control flow, and manage memory.

Common Pitfalls

- Confusing direct and indirect addressing, especially regarding the number of memory accesses.

Standard Problem-Solving Techniques

- Trace the instruction execution step-by-step to calculate the effective address for given register/memory contents.

Average Memory Access Time (AMAT)

AMAT is the average time taken to access memory, considering the presence of a cache and its hit/miss rates. It quantifies the performance of the memory hierarchy.

Important Formulas & Concepts

- For a single-level cache:

$$AMAT = T_{hit} + MR \times T_{miss_penalty}$$

where T_{hit} is cache hit time, MR is miss rate, and $T_{miss_penalty}$ is the time to fetch data from the next level of memory (main memory).

- For a two-level cache (L1, L2):

$$AMAT = T_{L1_hit} + MR_{L1} \times (T_{L2_hit} + MR_{L2} \times T_{main_memory})$$

where T_{L1_hit} is L1 hit time, MR_{L1} is L1 miss rate, T_{L2_hit} is L2 hit time, MR_{L2} is L2 miss rate (local miss rate for L2), and T_{main_memory} is main memory access time.

Key Properties & Identities

- Lower AMAT indicates better memory system performance.
- Affected by cache hit rate, miss penalty, and cache access times.

Common Pitfalls

- Forgetting to add the hit time to the miss penalty when calculating total miss time.
- Incorrectly applying local vs. global miss rates for multi-level caches.

Standard Problem-Solving Techniques

- Draw the memory hierarchy and clearly identify access times and miss rates for each level.

Bit Vector

A bit vector (or bit array) is a data structure that efficiently stores a sequence of bits. Each bit typically represents a boolean flag or the presence/absence of an element in a set.

Important Formulas & Concepts

- Storage: N bits require $\lceil N/8 \rceil$ bytes of storage.
- Bitwise operations: AND, OR, XOR, NOT, SHIFT for efficient manipulation.

Key Properties & Identities

- Space-efficient for large sets of boolean flags.
- Fast bitwise operations for checking, setting, or clearing flags.

Common Pitfalls

- Misinterpreting bit positions or the effect of bitwise operations.

Standard Problem-Solving Techniques

- Use bit masks to isolate or modify specific bits.

CISC RISC Architecture

CISC (Complex Instruction Set Computer) and RISC (Reduced Instruction Set Computer) are two contrasting philosophies for designing instruction sets. They represent different approaches to instruction complexity and hardware implementation.

Important Formulas & Concepts

- CISC Characteristics:**

- Many complex instructions, often performing multiple operations (e.g., memory access and arithmetic).
- Variable instruction length.
- Microprogrammed control unit.
- Fewer general-purpose registers, more memory-to-memory operations.
- Complex addressing modes.
- **RISC Characteristics:**
 - Few, simple, fixed-length instructions, typically one operation per instruction.
 - Hardwired control unit.
 - Many general-purpose registers, load/store architecture (only LOAD/STORE access memory).
 - Simple addressing modes.
 - Optimized for pipelining.

Key Properties & Identities

- RISC generally achieves higher CPI (Cycles Per Instruction) but lower clock cycle time due to simpler instructions, leading to better performance in many cases.
- CISC aims for fewer instructions per program but higher CPI.

Common Pitfalls

- Assuming one architecture is universally superior; their suitability depends on the application and compiler technology.

Standard Problem-Solving Techniques

- Compare and contrast features to identify the type of architecture described in a scenario.

Cache Memory

Cache memory is a small, fast memory component that stores copies of data from frequently used main memory locations. Its purpose is to reduce the average time to access data, exploiting the principle of locality.

Important Formulas & Concepts

- **Address Breakdown:** For a memory address of M bits, a cache with B bytes per block, and C cache lines (or sets):
 - **Block Offset bits:** $\log_2 B$
 - **Number of Blocks in Main Memory:** $2^M / B$
 - **Number of Cache Lines:** C
- **Direct Mapping:**
 - **Index bits:** $\log_2 C$
 - **Tag bits:** $M - (\log_2 C + \log_2 B)$
 - $Cache_Line_Index = (Main_Memory_Block_Number) \pmod{C}$
- **Set-Associative Mapping (S-way):**
 - **Number of Sets:** C/S
 - **Index bits:** $\log_2(C/S)$
 - **Tag bits:** $M - (\log_2(C/S) + \log_2 B)$
 - $Cache_Set_Index = (Main_Memory_Block_Number) \pmod{(C/S)}$
- **Fully Associative Mapping:**
 - No Index bits.
 - **Tag bits:** $M - \log_2 B$
 - Any block can go into any cache line.
- **Hit Rate (HR):** $\frac{\text{Number of Cache Hits}}{\text{Total Memory Accesses}}$
- **Miss Rate (MR):** $1 - HR$

Key Properties & Identities

- Exploits temporal and spatial locality of reference.
- Mapping techniques (Direct, Set-Associative, Fully Associative) determine where a block can be placed.
- Write policies (Write-Through, Write-Back) determine when data is written to main memory.

- Replacement policies (LRU, FIFO, LFU, Random) determine which block to evict on a miss.

Common Pitfalls

- Incorrectly calculating the number of bits for Tag, Index, and Offset.
- Confusing the total cache size with the number of cache lines.

Standard Problem-Solving Techniques

- Draw the cache structure and trace memory accesses to determine hits/misses and cache contents.

Conflict Misses

Conflict misses occur in direct-mapped or set-associative caches when multiple memory blocks map to the same cache line or set, and these blocks are actively used, causing them to evict each other even if other cache lines are empty.

Important Formulas & Concepts

- These are a type of "capacity miss" specifically due to the mapping function.
- Increasing associativity reduces conflict misses.

Key Properties & Identities

- Occur when the cache is not full but the required block is evicted due to mapping constraints.
- Distinguished from compulsory (first-time access) and capacity (cache too small) misses.

Common Pitfalls

- Mistaking a conflict miss for a capacity miss or a compulsory miss.

Standard Problem-Solving Techniques

- Analyze the memory access pattern and the cache mapping scheme to identify if multiple active blocks map to the same location.

Control Unit

The control unit is the part of the CPU that directs the operation of the processor. It interprets instructions and generates control signals to coordinate the activities of other components, such as the ALU, registers, and memory.

Important Formulas & Concepts

- **Hardwired Control:** Implemented using combinational logic gates.
 - Faster execution.
 - Complex to design and modify for large instruction sets.
 - Used in RISC processors.
- **Microprogrammed Control:** Implemented using a sequence of microinstructions stored in a control memory.
 - More flexible and easier to design/modify for complex instruction sets.
 - Slower execution due to memory access for microinstructions.
 - Used in CISC processors.

Key Properties & Identities

- Determines the sequence of micro-operations for each machine instruction.
- Generates timing and control signals for the entire system.

Common Pitfalls

- Not understanding the trade-offs between hardwired and microprogrammed control in terms of speed, flexibility, and design complexity.

Standard Problem-Solving Techniques

- Relate the control unit type to the characteristics of CISC/RISC architectures.

DMA (Direct Memory Access)

DMA is a hardware feature that allows I/O devices to transfer data directly to and from main memory without involving the CPU. This significantly reduces CPU overhead for data transfers, improving system performance.

Important Formulas & Concepts

- DMA controller manages the data transfer.
- **Cycle Stealing:** DMA controller takes control of the bus for one or more memory cycles to transfer data.
- **Burst Mode:** DMA controller takes control of the bus and transfers an entire block of data before releasing the bus.

Key Properties & Identities

- Frees the CPU to perform other tasks during I/O operations.
- Requires a DMA controller chip.

Common Pitfalls

- Misunderstanding that DMA completely bypasses the CPU; the CPU initiates and configures the DMA transfer.

Standard Problem-Solving Techniques

- Analyze scenarios to determine when DMA is most beneficial compared to programmed I/O or interrupt-driven I/O.

DRAM (Dynamic Random Access Memory)

DRAM is the most common type of main memory used in computers. It stores each bit of data in a separate capacitor within an integrated circuit, requiring periodic refreshing to maintain the charge and thus the data.

Important Formulas & Concepts

- Data stored as charge in capacitors.
- Requires refresh cycles to prevent data loss.

Key Properties & Identities

- Volatile (data is lost when power is removed).
- Higher density and lower cost per bit compared to SRAM.
- Slower access times than SRAM due to refresh cycles and capacitor charging/discharging.

Common Pitfalls

- Confusing DRAM characteristics with SRAM (Static RAM), which uses latches and does not require refreshing.

Standard Problem-Solving Techniques

- Compare and contrast DRAM with SRAM based on cost, speed, density, and power consumption.

Data Dependency

A data dependency exists between two instructions when the second instruction requires data produced by the first instruction, or when they both access the same memory location, potentially leading to incorrect results if executed out of order.

Important Formulas & Concepts

- **RAW (Read After Write) / True Dependency:** Instruction J tries to read a source before instruction I writes to it. $I \rightarrow J$. This is the most common and critical dependency.
- **WAR (Write After Read) / Anti-dependency:** Instruction J tries to write to a destination before instruction I reads from it. $I \leftarrow J$. Can be resolved by renaming.
- **WAW (Write After Write) / Output Dependency:** Instruction J tries to write to a destination before instruction I writes to it. $I \Rightarrow J$. Can be resolved by renaming.

Key Properties & Identities

- RAW dependencies are true data dependencies and must be preserved.
- WAR and WAW are name dependencies and can be eliminated by register renaming.

Common Pitfalls

- Incorrectly identifying the type of dependency or missing a dependency.

Standard Problem-Solving Techniques

- Analyze the read/write sets of instructions to identify dependencies.

Data Hazards

Data hazards occur in pipelined processors when an instruction attempts to access data before a preceding instruction has made that data available. These hazards can lead to incorrect program execution if not handled.

Important Formulas & Concepts

- Caused by RAW, WAR, and WAW dependencies.
- **Solutions:**
 - **Stalling (Bubbles/NOPs):** Insert NOP instructions to delay the dependent instruction.
 - **Forwarding (Bypassing):** Route the result of an instruction directly from an internal pipeline register to the input of a subsequent instruction's functional unit, bypassing the register file.
 - **Compiler Scheduling:** Reorder instructions at compile time to avoid hazards.

Key Properties & Identities

- Increase the CPI (Cycles Per Instruction) of a pipeline, reducing ideal speedup.
- Forwarding is the most common hardware solution to reduce stalls due to RAW hazards.

Common Pitfalls

- Calculating the number of stalls incorrectly, especially when forwarding is present.
- Not identifying all data dependencies in a given instruction sequence.

Standard Problem-Solving Techniques

- Draw pipeline diagrams to visualize instruction flow and identify where stalls or forwarding are needed.

Data Path

The data path of a CPU consists of the functional units (like the ALU, registers, and multiplexers) and the buses that connect them, through which data flows during instruction execution. It performs the actual data processing operations.

Important Formulas & Concepts

- Key components: Program Counter (PC), Instruction Register (IR), Memory Address Register (MAR), Memory Data Register (MDR), General Purpose Registers (GPRs), Arithmetic Logic Unit (ALU).

Key Properties & Identities

- The data path is controlled by signals generated by the control unit.
- Its design impacts the clock cycle time and the number of cycles per instruction.

Common Pitfalls

- Not understanding how data moves between components for different instruction types.

Standard Problem-Solving Techniques

- Trace the flow of data for a specific instruction (e.g., LOAD, ADD) through the datapath components.

Direct Mapping

Direct mapping is a cache organization technique where each block from main memory can only be placed into one specific line in the cache. This mapping is determined by a simple modular arithmetic function of the memory block address.

Important Formulas & Concepts

- **Cache Line Index:**

$$\text{Cache_Line_Index} = (\text{Main_Memory_Block_Number}) \pmod{\text{Number_of_Cache_Lines}}$$
- **Address Breakdown:** A memory address is divided into three fields: Tag, Index, and Block Offset.
 - **Block Offset bits:** $\log_2(\text{Block Size in bytes})$
 - **Index bits:** $\log_2(\text{Number of Cache Lines})$
 - **Tag bits:** Total Address Bits – Index Bits – Block Offset Bits

Key Properties & Identities

- Simplest to implement and fastest for cache lookup.
- Suffers from high conflict misses if frequently accessed blocks map to the same cache line.

Common Pitfalls

- Incorrectly calculating the number of bits for the Tag, Index, or Block Offset fields.

Standard Problem-Solving Techniques

- Given memory and cache parameters, determine the address breakdown and trace memory accesses to identify hits/misses.

Dirty Bit

A dirty bit (or modified bit) is a flag associated with a cache line or a virtual memory page table entry. It indicates whether the corresponding data block in the cache/page has been modified since it was loaded from the next lower level of memory (main memory/disk).

Important Formulas & Concepts

- Set to '1' when data in the cache line or page is written to.
- Set to '0' when the data is clean (matches the lower level memory).

Key Properties & Identities

- Crucial for write-back cache policies: if the dirty bit is set, the block must be written back to main memory upon eviction.
- Used in virtual memory page replacement algorithms to avoid unnecessary writes to disk.

Common Pitfalls

- Misunderstanding its role in write-through caches (where it's not typically needed) versus write-back caches.

Standard Problem-Solving Techniques

- Analyze cache write operations and page replacement scenarios to determine when the dirty bit is set or checked.

Disk

A disk (hard disk drive) is a non-volatile secondary storage device that stores data magnetically on rotating platters. It is characterized by its large capacity and persistence, but significantly slower access times compared to RAM.

Important Formulas & Concepts

- **Disk Access Time:**

$$Access_Time = Seek_Time + Rotational_Latency + Transfer_Time$$

- **Seek Time:** Time taken for the read/write head to move to the correct track. (Often given as average or calculated based on number of tracks).
- **Rotational Latency:** Time taken for the desired sector to rotate under the read/write head.

$$Average_Rotational_Latency = \frac{1}{2} \times \frac{60 \text{ seconds}}{\text{RPM}}$$

(where RPM is revolutions per minute).

- **Transfer Time:** Time taken to transfer the actual data once the head is positioned.

$$Transfer_Time = \frac{\text{Number of Sectors to Transfer}}{\text{Sectors per Track}} \times \frac{60 \text{ seconds}}{\text{RPM}}$$

$$\text{or } Transfer_Time = \frac{\text{Amount of Data}}{\text{Transfer Rate}}.$$

Key Properties & Identities

- Non-volatile storage.
- Mechanical components make it orders of magnitude slower than electronic memory.

Common Pitfalls

- Incorrectly calculating average rotational latency (using 1/2 of a full rotation).
- Mixing units (e.g., milliseconds and seconds).

Standard Problem-Solving Techniques

- Break down the access time calculation into its three components and calculate each separately.

Hazards

Hazards are situations in pipelined processors that prevent the next instruction in the instruction stream from executing in its designated clock cycle. They reduce the ideal speedup of a pipeline.

Important Formulas & Concepts

- **Structural Hazards:** Occur when two instructions require the same hardware resource at the same time (e.g., single memory port for both instruction fetch and data access).
- **Data Hazards:** Occur when an instruction depends on the result of a previous instruction that has not yet completed (RAW, WAR, WAW).
- **Control Hazards (Branch Hazards):** Occur when the pipeline makes a wrong guess about the next instruction to fetch (e.g., after a conditional branch).

Key Properties & Identities

- All hazards increase the effective CPI (Cycles Per Instruction) of the pipeline.

- Solutions include stalling, forwarding, and branch prediction.

Common Pitfalls

- Confusing the different types of hazards or misidentifying the cause of a stall.

Standard Problem-Solving Techniques

- Analyze the instruction sequence and pipeline stage usage to identify potential conflicts.

IO Handling

I/O handling refers to the mechanisms by which a CPU communicates and exchanges data with peripheral input/output devices. Efficient I/O handling is crucial for system performance.

Important Formulas & Concepts

- **Programmed I/O (Polling):** CPU continuously checks the status of an I/O device.
 - High CPU overhead, simple to implement.
- **Interrupt-Driven I/O:** I/O device signals the CPU via an interrupt when it needs attention.
 - Lower CPU overhead than polling, CPU can do other work.
 - Requires interrupt service routines (ISRs).
- **Direct Memory Access (DMA):** I/O device transfers data directly to/from memory without CPU intervention.
 - Lowest CPU overhead for large data transfers.
 - Requires a DMA controller.

Key Properties & Identities

- Each method has trade-offs in terms of CPU utilization, response time, and complexity.
- Interrupts allow for asynchronous event handling.

Common Pitfalls

- Not understanding the CPU's involvement (or lack thereof) in data transfer for each method.

Standard Problem-Solving Techniques

- Evaluate scenarios to determine the most suitable I/O handling method based on data volume and CPU availability.

Instruction Execution

Instruction execution is the process by which a CPU carries out a single machine instruction. In a pipelined processor, this process is broken down into several sequential stages, allowing multiple instructions to be processed concurrently.

Important Formulas & Concepts

- **Typical 5-stage pipeline:**
 1. **IF (Instruction Fetch):** Fetch instruction from memory.
 2. **ID (Instruction Decode):** Decode instruction, read registers.
 3. **EX (Execute):** Perform ALU operation.
 4. **MEM (Memory Access):** Access data memory (load/store).
 5. **WB (Write Back):** Write result back to register file.

Key Properties & Identities

- Each stage ideally takes one clock cycle in a pipelined processor.
- The slowest stage determines the clock cycle time of the pipeline.

Common Pitfalls

- Misunderstanding the specific function of each pipeline stage.

Standard Problem-Solving Techniques

- Trace an instruction through each stage of the pipeline to understand its execution flow.

Instruction Format

The instruction format defines the layout of bits within a machine instruction. It specifies how the opcode, operands, and addressing mode information are encoded, directly impacting the instruction set's flexibility and the CPU's decoding complexity.

Important Formulas & Concepts

- An instruction typically consists of an **opcode** (operation code) and one or more **operand fields**.
- Operand fields can specify registers, memory addresses, or immediate values.
- **Fixed-length instruction format:** All instructions have the same length. Simplifies fetching and decoding, good for pipelining (RISC).
- **Variable-length instruction format:** Instructions can have different lengths. Allows for more complex instructions and addressing modes, but complicates fetching and decoding (CISC).

Key Properties & Identities

- The number of bits for the opcode determines the maximum number of unique operations.
- The number of bits for operand fields determines the number of registers or the addressable memory range.

Common Pitfalls

- Incorrectly calculating the maximum number of opcodes or addressable memory given a fixed instruction length.

Standard Problem-Solving Techniques

- Given instruction length and operand requirements, calculate the remaining bits for the opcode or vice-versa.

Instruction Set Architecture (ISA)

The ISA is the abstract model of a computer that is visible to a programmer or compiler writer. It defines the set of instructions, registers, memory organization, data types, and the I/O model that the hardware supports.

Important Formulas & Concepts

- Includes the instruction set, register set, memory addressing modes, and interrupt handling mechanisms.
- Forms the interface between software and hardware.

Key Properties & Identities

- ISA defines *what* the processor does, while microarchitecture defines *how* it does it.
- CISC and RISC are two major ISA design philosophies.

Common Pitfalls

- Confusing ISA (the specification) with microarchitecture (the implementation).

Standard Problem-Solving Techniques

- Understand that ISA is the contract that allows software to run on different hardware implementations of the same ISA.

Interrupts

An interrupt is a signal to the CPU indicating an event that requires immediate attention, causing the CPU to temporarily suspend its current task and execute a special routine (Interrupt Service Routine or ISR) to handle the event.

Important Formulas & Concepts

- **Hardware Interrupts:** Generated by I/O devices (e.g., keyboard press, disk completion). Asynchronous.
- **Software Interrupts (Exceptions/Traps):** Generated by program execution (e.g., division by zero, page fault, system call). Synchronous.
- **Interrupt Vector Table:** A table of addresses for ISRs, indexed by interrupt number.

Key Properties & Identities

- Enable efficient I/O handling and error management without constant CPU polling.
- Involve saving CPU state, executing ISR, and restoring CPU state.

Common Pitfalls

- Misunderstanding the sequence of events during interrupt handling (saving context, jumping to ISR, restoring context).

Standard Problem-Solving Techniques

- Trace the flow of control when an interrupt occurs, considering context saving and ISR execution.

Machine Instruction

A machine instruction is a command or operation that a CPU can directly understand and execute. It is represented in binary code and forms the lowest-level programming language, specific to a particular processor's Instruction Set Architecture.

Important Formulas & Concepts

- Composed of an opcode and zero or more operands.
- Each instruction performs a basic operation (e.g., add, load, store, branch).

Key Properties & Identities

- Directly executable by the hardware.
- The fundamental unit of computation for the CPU.

Common Pitfalls

- Confusing machine instructions (binary) with assembly language instructions (mnemonic representation).

Standard Problem-Solving Techniques

- Understand how an assembly instruction translates into its binary machine code equivalent based on the instruction format.

Memory Interfacing

Memory interfacing is the process of connecting memory devices (RAM, ROM) to the CPU. It involves designing the necessary address decoding logic, connecting data and address buses, and generating control signals to ensure proper communication and data transfer.

Important Formulas & Concepts

- **Address Decoding:** Logic gates (decoders, AND gates) used to select the correct memory chip based on the CPU's address lines.
- **Chip Select (CS):** A control signal that enables a specific memory chip.

- **Memory Map:** The allocation of address ranges to different memory devices.

Key Properties & Identities

- Ensures that each memory location has a unique address and that the correct device responds to a CPU request.
- Bus width (data and address) must match between CPU and memory.

Common Pitfalls

- Incorrectly designing address decoding logic, leading to address conflicts or unused address space.

Standard Problem-Solving Techniques

- Given CPU address lines and memory chip sizes, determine the address ranges and design the minimal decoding logic.

Microprogramming

Microprogramming is a technique for implementing the control unit of a CPU by storing a sequence of microinstructions in a special control memory. Each machine instruction is executed by a microprogram, which is a sequence of microinstructions.

Important Formulas & Concepts

- **Microinstruction:** A low-level instruction that controls the individual functional units of the CPU (e.g., enable ALU, load register).
- **Control Store (CS):** Memory where microprograms are stored.

Key Properties & Identities

- Offers flexibility: easier to design complex instruction sets and modify existing ones.
- Slower than hardwired control due to the overhead of fetching and decoding microinstructions.
- Commonly used in CISC processors.

Common Pitfalls

- Confusing microinstructions (internal CPU control) with machine instructions (user-level instructions).

Standard Problem-Solving Techniques

- Understand how a machine instruction is broken down into a sequence of micro-operations controlled by microinstructions.

Pipelining

Pipelining is a technique that allows multiple instructions to be in different stages of execution simultaneously, improving the throughput of a processor. It does not reduce the execution time of a single instruction but increases the rate at which instructions complete.

Important Formulas & Concepts

- **Number of stages:** K
- **Number of instructions:** N
- **Clock cycle time:** T_{cycle} (determined by the slowest stage).
- **Execution time for N instructions (non-pipelined):** $T_{non-pipeline} = N \times K \times T_{cycle}$
- **Execution time for N instructions (ideal pipelined):** $T_{pipeline} = (K + N - 1) \times T_{cycle}$
- **Ideal Speedup (S) for large N:** $S = \frac{N \times K \times T_{cycle}}{(K + N - 1) \times T_{cycle}} \approx K$
- **Throughput (ideal):** $\frac{1}{T_{cycle}}$ instructions per cycle.
- **CPI (Cycles Per Instruction):**

- Ideal CPI = 1.
- Actual CPI = $1 + \text{Stalls per instruction}$.

Key Properties & Identities

- Increases instruction throughput, not latency.
- Performance is limited by hazards (structural, data, control).

Common Pitfalls

- Incorrectly calculating execution time when stalls are present.
- Assuming pipelining reduces the execution time of a single instruction.

Standard Problem-Solving Techniques

- Draw pipeline diagrams to visualize instruction flow, identify hazards, and calculate stalls.

Runtime Environment

The runtime environment refers to the state of a program during its execution, encompassing all resources and conditions necessary for the program to run. This includes the CPU's registers, memory (stack, heap, code, data segments), program counter, and operating system services.

Important Formulas & Concepts

- Includes CPU state (registers, PC, flags), memory layout (code, data, heap, stack), and OS resources.

Key Properties & Identities

- Managed by the operating system and supported by the hardware architecture.
- Defines how programs interact with the underlying system.

Common Pitfalls

- Not understanding the distinction between compile-time and run-time aspects of a program.

Standard Problem-Solving Techniques

- Understand how memory is organized and utilized by a program during execution (e.g., stack for function calls, heap for dynamic allocation).

Speedup

Speedup is a measure of how much faster a task executes on an improved system compared to an original system. It quantifies the performance gain achieved by optimizations like pipelining, parallel processing, or using faster components.

Important Formulas & Concepts

- **General Speedup:**

$$\text{Speedup} = \frac{\text{Execution Time}_{\text{old}}}{\text{Execution Time}_{\text{new}}}$$

- **Amdahl's Law:** Calculates the theoretical maximum speedup for a system when only a portion of the task can be improved.

$$S_{\text{overall}} = \frac{1}{(1 - P) + \frac{P}{S_{\text{enhanced}}}}$$

where P is the proportion of the program that can be enhanced, and $S_{enhanced}$ is the speedup of the enhanced part.

- **Pipelining Speedup (for large N):** $S \approx K$ (number of pipeline stages).

Key Properties & Identities

- Amdahl's Law highlights that the sequential portion of a program limits overall speedup.
- Speedup is a dimensionless quantity.

Common Pitfalls

- Incorrectly applying Amdahl's Law, especially regarding the value of P .
- Forgetting to account for overheads or non-ideal conditions in pipelining.

Standard Problem-Solving Techniques

- Identify the parallelizable and sequential components of a task when using Amdahl's Law.

Stall

A stall (also known as a bubble or NOP) is a delay introduced into a pipeline to resolve a hazard. When a hazard is detected, the pipeline is temporarily halted for one or more clock cycles, preventing the dependent instruction from proceeding until the hazard is cleared.

Important Formulas & Concepts

- Stalls increase the CPI (Cycles Per Instruction) of the pipeline.
- Number of stalls depends on the type of hazard and the pipeline's ability to forward data.

Key Properties & Identities

- Reduces the effective throughput and speedup of the pipeline.
- A common method to ensure correctness in the presence of data and control hazards.

Common Pitfalls

- Incorrectly calculating the number of stalls required for a given hazard, especially when forwarding is implemented.

Standard Problem-Solving Techniques

- Draw pipeline diagrams and mark the clock cycles where stalls are inserted to resolve dependencies.

Virtual Memory

Virtual memory is a memory management technique that allows a computer to compensate for physical memory shortages by temporarily transferring data from RAM to disk storage. It provides the illusion of a larger, contiguous memory space to programs.

Important Formulas & Concepts

- **Paging:** Divides memory into fixed-size blocks (pages and frames).
- **Page Table:** Maps virtual page numbers to physical frame numbers.
- **TLB (Translation Lookaside Buffer):** A small, fast cache for page table entries.
- **Effective Access Time (EAT) with TLB:**

$$EAT = TLB_{hit_rate} \times (T_{TLB} + T_{memory}) + TLB_{miss_rate} \times (T_{TLB} + 2 \times T_{memory})$$

where T_{TLB} is TLB access time and T_{memory} is main memory access time.

- **Effective Access Time (EAT) with Page Faults:**

$$EAT = (1 - P) \times T_{memory} + P \times T_{page_fault}$$

where P is the page fault rate, T_{memory} is memory access time, and T_{page_fault} is the time to handle a page fault (including disk I/O).

Key Properties & Identities

- Provides memory protection, allows multitasking, and enables programs larger than physical memory.
- Address translation involves converting virtual addresses to physical addresses.

Common Pitfalls

- Incorrectly calculating EAT, especially when considering both TLB misses and page faults.
- Confusing page table entries with TLB entries.

Standard Problem-Solving Techniques

- Trace the address translation process (TLB lookup, page table lookup, page fault handling).

Quick Formula Reference

Topic	Formula/Concept
Average Memory Access Time (AMAT)	• Single-level cache: $AMAT = T_{hit} + MR \times T_{miss_penalty}$ • Two-level cache: $AMAT = T_{L1_hit} + MR_{L1} \times (T_{L2_hit} + MR_{L2} \times T_{main_memory})$
Cache Memory (Address Breakdown)	• Block Offset bits: $\log_2(\text{Block Size})$ • Direct Mapped Index bits: $\log_2(\text{Number of Cache Lines})$ • Set-Associative Index bits: $\log_2(\text{Number of Sets})$ • Tag bits: Total Address Bits – Index Bits – Block Offset Bits • Direct Mapped Index: $(\text{Main Memory Block Number}) \pmod{\text{Number of Cache Lines}}$
Disk Access Time	• $Access_Time = Seek_Time + Rotational_Latency + Transfer_Time$ • $Average_Rotational_Latency = \frac{1}{2} \times \frac{60}{RPM}$ • $Transfer_Time = \frac{\text{Number of Sectors}}{\text{Sectors per Track}} \times \frac{60}{RPM}$
Pipelining Performance	• Execution time (N instructions, K stages, ideal): $T_{pipeline} = (K + N - 1) \times T_{cycle}$ • Ideal Speedup (large N): $S \approx K$ • Actual CPI: $1 + \text{Stalls per instruction}$
Speedup (Amdahl's Law)	• $S_{overall} = \frac{1}{(1-P) + \frac{P}{S_{enhanced}}}$
Virtual Memory (EAT)	• With TLB: $EAT = TLB_{hit_rate} \times (T_{TLB} + T_{memory}) + TLB_{miss_rate} \times (T_{TLB} + 2 \times T_{memory})$ • With Page Faults: $EAT = (1 - P) \times T_{memory} + P \times T_{page_fault}$

Important Tips for GATE

- **Master Numerical Problems:** A significant portion of COA questions are numerical, especially on Cache Memory (AMAT, address breakdown), Pipelining (execution time, speedup, stalls), Disk Access Time, and Virtual Memory (EAT). Practice these extensively.
- **Address Breakdown is Key:** For cache and virtual memory, thoroughly understand how a logical/physical address is divided into Tag, Index/Page Number, and Offset. This is fundamental for many problems.
- **Pipeline Diagrams:** Learn to draw and analyze pipeline diagrams for instruction sequences. This is the best way to identify data/control hazards and calculate the number of stalls or benefits of forwarding.

- **Conceptual Clarity:** Don't just memorize definitions. Understand the "why" behind concepts. For example, why RISC is better for pipelining, why cache works (locality), or why DMA is used.
- **Distinguish Similar Concepts:** Be clear on the differences between CISC/RISC, hardwired/microprogrammed control, various addressing modes, and different types of hazards (structural, data, control).
- **Read Questions Carefully:** Pay close attention to units (ns, ms, cycles), specific assumptions (e.g., "no forwarding," "average rotational latency"), and what is being asked (e.g., total time vs. speedup).
- **Amdahl's Law:** This formula frequently appears in questions related to performance improvement. Understand how to apply it correctly, especially identifying the parallelizable portion P .
- **Time Management:** Some numerical problems can be lengthy. If you get stuck or find a calculation too complex, consider moving on and returning later. Look for shortcuts or approximations if applicable.

1.1

Addressing Modes (19)

Practice Test: Test 1 (15Q)

1.1.1 Addressing Modes: GATE CSE 1987 | Question: 1-V

The most relevant addressing mode to write position-independent codes is:

A. Direct mode B. Indirect mode C. Relative mode D. Indexed mode

gate1987 co-and-architecture addressing-modes easy

Answer key

1.1.2 Addressing Modes: GATE CSE 1988 | Question: 9iii

In the program scheme given below indicate the instructions containing any operand needing relocation for position independent behaviour. Justify your answer.

```

Y = 10
MOV    X(R0), R1
MOV    X, R0
MOV    2(R0), R1
MOV    Y(R0), R5
.
.
.
X:  WORD  0,0,0

```

gate1988 normal descriptive co-and-architecture addressing-modes

Answer key

1.1.3 Addressing Modes: GATE CSE 1989 | Question: 2-ii

Match the pairs in the following questions:

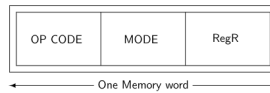
(A) Base addressing	(p) Reentranecy
(B) Indexed addressing	(q) Accumulator
(C) Stack addressing	(r) Array
(D) Implied addressing	(s) Position independent

gate1989 match-the-following co-and-architecture addressing-modes easy

Answer key

1.1.4 Addressing Modes: GATE CSE 1993 | Question: 10

The instruction format of a CPU is:



Mode and RegR together specify the operand. RegR specifies a CPU register and Mode specifies an addressing mode. In particular, Mode = 2 specifies that 'the register RegR contains the address of the operand, after fetching the operand, the contents of RegR are incremented by 1'.

An instruction at memory location 2000 specifies Mode = 2 and the RegR refers to program counter (PC).

- A. What is the address of the operand?
- B. Assuming that is a non-jump instruction, what are the contents of PC after the execution of this instruction?

gate1993 co-and-architecture addressing-modes normal descriptive

Answer key

1.1.5 Addressing Modes: GATE CSE 1996 | Question: 1.16, ISRO2016-42



Relative mode of addressing is most relevant to writing:

- A. Co – routines
- B. Position – independent code
- C. Shareable code
- D. Interrupt Handlers

gate1996 co-and-architecture addressing-modes easy isro2016

Answer key

1.1.6 Addressing Modes: GATE CSE 1998 | Question: 1.19



Which of the following addressing modes permits relocation without any change whatsoever in the code?

- A. Indirect addressing
- B. Indexed addressing
- C. Base register addressing
- D. PC relative addressing

gate1998 co-and-architecture addressing-modes easy

Answer key

1.1.7 Addressing Modes: GATE CSE 1999 | Question: 2.23



A certain processor supports only the immediate and the direct addressing modes. Which of the following programming language features cannot be implemented on this processor?

- A. Pointers
- B. Arrays
- C. Records
- D. Recursive procedures with local variable

gate1999 co-and-architecture addressing-modes normal multiple-selects

Answer key

1.1.8 Addressing Modes: GATE CSE 2000 | Question: 1.10



The most appropriate matching for the following pairs

- | | |
|------------------------------|--------------|
| X: Indirect addressing | 1: Loops |
| Y: Immediate addressing | 2: Pointers |
| Z: Auto decrement addressing | 3: Constants |

is

- A. $X - 3, Y - 2, Z - 1$
- B. $X - 1, Y - 3, Z - 2$
- C. $X - 2, Y - 3, Z - 1$
- D. $X - 3, Y - 1, Z - 2$

gatecse-2000 co-and-architecture easy addressing-modes match-the-following

Answer key

1.1.9 Addressing Modes: GATE CSE 2001 | Question: 2.9



Which is the most appropriate match for the items in the first column with the items in the second column:

X. Indirect Addressing	I. Array implementation
Y. Indexed Addressing	II. Writing relocatable code
Z. Base Register Addressing	III. Passing array as parameter

- A. (X, III), (Y, I), (Z, II)
C. (X, III), (Y, II), (Z, I)

- B. (X, II), (Y, III), (Z, I)
D. (X, I), (Y, III), (Z, II)

gatecse-2001 co-and-architecture addressing-modes easy match-the-following

Answer key

1.1.10 Addressing Modes: GATE CSE 2002 | Question: 1.24



In the absolute addressing mode:

- A. the operand is inside the instruction
B. the address of the operand is inside the instruction
C. the register containing the address of the operand is specified inside the instruction
D. the location of the operand is implicit

gatecse-2002 co-and-architecture addressing-modes easy

Answer key

1.1.11 Addressing Modes: GATE CSE 2004 | Question: 20



Which of the following addressing modes are suitable for program relocation at run time?

- I. Absolute addressing
II. Based addressing
III. Relative addressing
IV. Indirect addressing

- A. I and IV
C. II and III

- B. I and II
D. I, II and IV

gatecse-2004 co-and-architecture addressing-modes easy

Answer key

1.1.12 Addressing Modes: GATE CSE 2005 | Question: 65



Consider a three word machine instruction

$ADDA[R_0], @B$

The first operand (destination) " $A[R_0]$ " uses indexed addressing mode with R_0 as the index register. The second operand (source) " $@B$ " uses indirect addressing mode. A and B are memory addresses residing at the second and third words, respectively. The first word of the instruction specifies the opcode, the index register designation and the source and destination addressing modes. During execution of **ADD** instruction, the two operands are added and stored in the destination (first operand).

The number of memory cycles needed during the execution cycle of the instruction is:

- A. 3 B. 4 C. 5 D. 6

gatecse-2005 co-and-architecture addressing-modes normal

Answer key

1.1.13 Addressing Modes: GATE CSE 2005 | Question: 66



Match each of the high level language statements given on the left hand side with the most natural addressing mode from those listed on the right hand side.

- | | |
|--------------------------------|-------------------------|
| (1) $A[I] = B[J]$ | (a) Indirect addressing |
| (2) <code>while (*A++);</code> | (b) Indexed addressing |
| (3) <code>int temp = *x</code> | (c) Auto increment |

- A. $(1, c), (2, b), (3, a)$
 C. $(1, b), (2, c), (3, a)$

- B. $(1, c), (2, c), (3, b)$
 D. $(1, a), (2, b), (3, c)$

gatecse-2005 co-and-architecture addressing-modes match-the-following easy

Answer key

1.1.14 Addressing Modes: GATE CSE 2008 | Question: 33, ISRO2009-80



Which of the following is/are true of the auto-increment addressing mode?

- I. It is useful in creating self-relocating code
- II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation
- III. The amount of increment depends on the size of the data item accessed

- A. I only B. II only C. III only D. II and III only

gatecse-2008 addressing-modes co-and-architecture normal isro2009

Answer key

1.1.15 Addressing Modes: GATE CSE 2011 | Question: 21



Consider a hypothetical processor with an instruction of type `LW R1, 20(R2)`, which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of a constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for the operand in memory?

- | | |
|--|----------------------------|
| A. Immediate addressing | B. Register addressing |
| C. Register Indirect Scaled Addressing | D. Base Indexed Addressing |

gatecse-2011 co-and-architecture addressing-modes easy

Answer key

1.1.16 Addressing Modes: GATE CSE 2017 | Set 1 | Question: 11



Consider the *C* struct defined below:

```
struct data {
    int marks[100];
    char grade;
    int cnumber;
};
struct data student;
```

The base address of student is available in register R1. The field student.grade can be accessed efficiently using:

- A. Post-increment addressing mode, $(R1)++$
- B. Pre-decrement addressing mode, $--(R1)$
- C. Register direct addressing mode, R1
- D. Index addressing mode, $X(R1)$, where *X* is an offset represented in 2's complement 16-bit representation

gatecse-2017-set1 co-and-architecture addressing-modes

Answer key

1.1.17 Addressing Modes: GATE CSE 2026 | Set 1 | Question: 4



Match each addressing mode in **List I** with a data element or an element of a data structure (in a high-level language) in **List II**:

List I	List II
P. Immediate	1. Element of an array
Q. Indirect	2. Pointer
R. Base with index	3. Element of a record
S. Base with offset/displacement	4. Constant

- A. P – 4, Q – 3, R – 1, S – 2
 B. P – 4, Q – 2, R – 1, S – 3
 C. P – 1, Q – 4, R – 3, S – 2
 D. P – 2, Q – 3, R – 1, S – 4

gatecse-2026-set1 co-and-architecture addressing-modes one-mark

Answer key

1.1.18 Addressing Modes: GATE IT 2006 | Question: 39, ISRO2009-42



Which of the following statements about relative addressing mode is FALSE?

- A. It enables reduced instruction size
 B. It allows indexing of array element with same instruction
 C. It enables easy relocation of data
 D. It enables faster address calculation than absolute addressing

gateit-2006 co-and-architecture addressing-modes normal isro2009

Answer key

1.1.19 Addressing Modes: GATE IT 2006 | Question: 40



The memory locations 1000, 1001 and 1020 have data values 18, 1 and 16 respectively before the following program is executed.

```

MOVI    Rs, 1           ; Move immediate
LOAD    Rd, 1000(Rs)    ; Load from memory
ADDI    Rd, 1000        ; Add immediate
STOREI  0(Rd), 20       ; Store immediate

```

Which of the statements below is TRUE after the program is executed ?

- A. Memory location 1000 has value 20
 B. Memory location 1020 has value 20
 C. Memory location 1021 has value 20
 D. Memory location 1001 has value 20

gateit-2006 co-and-architecture addressing-modes normal

Answer key

1.2 Average Memory Access Time (3)

1.2.1 Average Memory Access Time: GATE CSE 1992 | Question: 5-a



The access times of the main memory and the Cache memory, in a computer system, are 500 n sec and 50 nsec, respectively. It is estimated that 80% of the main memory request are for read the rest for write. The hit ratio for the read access only is 0.9 and a write-through policy (where both main and cache memories are updated simultaneously) is used. Determine the average time of the main memory (in ns).

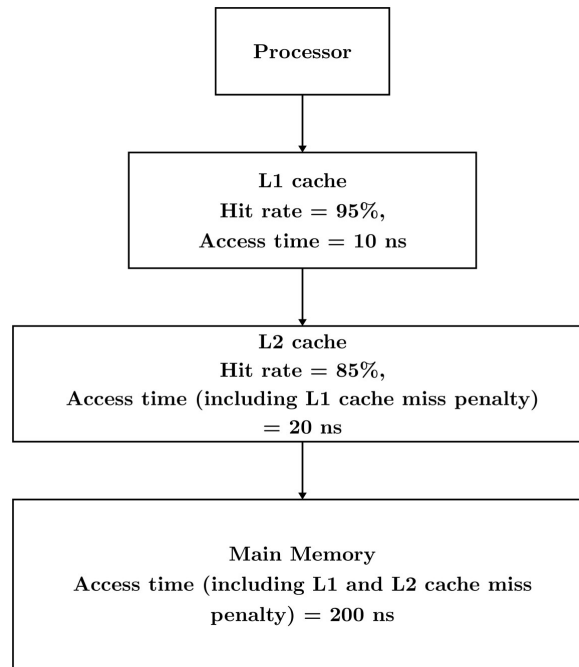
gate1992 co-and-architecture cache-memory normal numerical-answers average-memory-access-time

Answer key

1.2.2 Average Memory Access Time: GATE CSE 2025 | Set 1 | Question: 43



A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. Hit rates and access times of various memory units are shown in the figure. The average memory access time in nanoseconds (ns) is _____. (rounded off to two decimal places)



gatecse2025-set1 co-and-architecture cache-memory multilevel-cache average-memory-access-time numerical-answers two-marks

Answer key

1.2.3 Average Memory Access Time: GATE CSE 2025 | Set 2 | Question: 45



Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the "hit rate" for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 80% and the "miss penalty" for transferring data from L2 cache to L1 cache is 10 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 100 ns.

Then the average memory access time in this system in nanoseconds is _____. (rounded off to one decimal place)

gatecse2025-set2 co-and-architecture cache-memory average-memory-access-time multilevel-cache numerical-answers two-marks

Answer key

1.3 Bit Vector (1)

1.3.1 Bit Vector: GATE CSE 2026 | Set 2 | Question: 43



To keep track of free blocks in a file system, one of the two approaches is generally used - using bitmaps (bit vectors) or using linked lists. Consider that the linked list approach is used to keep track of free blocks in a file system. Assume that the disk size is 16 GB, block size is 2 KB, and block numbers used are 32-bit long. A single pointer of size 4 bytes is used in each block of the list to point to the next block of the list. The number of blocks required to hold the free disk block numbers is _____. (answer in integer)

Note: $1K = 2^{10}$ and $1G = 2^{30}$

gatecse-2026-set2 co-and-architecture bit-vector file-system numerical-answers two-marks

1.4.1 CISC RISC Architecture: GATE CSE 1999 | Question: 2.22



The main difference(s) between a CISC and a RISC processor is/are that a RISC processor typically

- A. has fewer instructions
- B. has fewer addressing modes
- C. has more registers
- D. is easier to implement using hard-wired logic

gate1999 co-and-architecture normal cisc-risc-architecture multiple-selects

1.4.2 CISC RISC Architecture: GATE CSE 2018 | Question: 5



Consider the following processor design characteristics:

- I. Register-to-register arithmetic operations only
- II. Fixed-length instruction format
- III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II and III

gatecse-2018 co-and-architecture cisc-risc-architecture easy one-mark

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(15Q\)](#) [Test 6 \(15Q\)](#) [Test 7 \(15Q\)](#) [Test 8 \(4Q\)](#)

1.5.1 Cache Memory: GATE CSE 1987 | Question: 4b



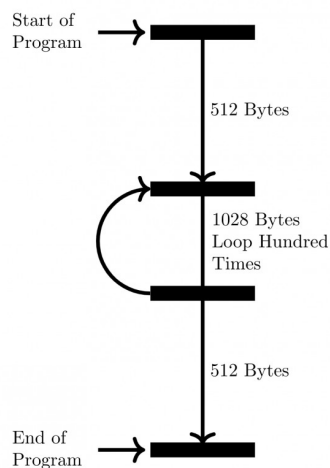
What is cache memory? What is rationale of using cache memory?

gate1987 co-and-architecture cache-memory descriptive

1.5.2 Cache Memory: GATE CSE 1989 | Question: 6a



A certain computer system was designed with cache memory of size 1 Kbytes and main memory size of 256 Kbytes. The cache implementation was fully associative cache with 4 bytes per block. The CPU memory data path was 16 bits and the memory was 2—way interleaved. Each memory read request presents two 16—bit words. A program with the model shown below was run to evaluate the cache design.



Answer the following questions:

- What is the hit ratio?
- Suggest a change in the program size of model to improve the hit ratio significantly.

gate1989 descriptive co-and-architecture cache-memory

Answer key

1.5.3 Cache Memory: GATE CSE 1990 | Question: 7a



A block-set associative cache memory consists of 128 blocks divided into four block sets. The main memory consists of 16,384 blocks and each block contains 256 eight bit words.

- How many bits are required for addressing the main memory?
- How many bits are needed to represent the TAG, SET and WORD fields?

gate1990 descriptive co-and-architecture cache-memory

Answer key

1.5.4 Cache Memory: GATE CSE 1993 | Question: 11



In the three-level memory hierarchy shown in the following table, p_i denotes the probability that an access request will refer to M_i .

Hierarchy Level (M_i)	Access Time (t_i)	Probability of Access (p_i)	Page Transfer Time (T_i)
M_1	10^{-6}	0.99000	0.001 sec
M_2	10^{-5}	0.00998	0.1 sec
M_3	10^{-4}	0.00002	---

If a miss occurs at level M_i , a page transfer occurs from M_{i+1} to M_i and the average time required for such a page swap is T_i . Calculate the average time t_A required for a processor to read one word from this memory system.

gate1993 co-and-architecture cache-memory normal descriptive

Answer key

1.5.5 Cache Memory: GATE CSE 1995 | Question: 1.6



The principle of locality justifies the use of:

- A. Interrupts B. DMA C. Polling D. Cache Memory

gate1995 co-and-architecture cache-memory easy

Answer key

1.5.6 Cache Memory: GATE CSE 1995 | Question: 2.25



A computer system has a 4 K word cache organized in block-set-associative manner with 4 blocks per set, 64 words per block. The number of bits in the SET and WORD fields of the main memory address format is:

- A. 15,40 B. 6,4 C. 7,2 D. 4,6

gate1995 co-and-architecture cache-memory normal

Answer key

1.5.7 Cache Memory: GATE CSE 1996 | Question: 26



A computer system has a three-level memory hierarchy, with access time and hit ratios as shown below:

Level 1 (Cache memory) Access time = $50nsec/byte$		Level 2 (Main memory) Access time = $200nsec/byte$		Level 3 Access time = $5\mu sec/byte$	
Size	Hit ratio	Size	Hit ratio	Size	Hit ratio
8M bytes	0.80	4M bytes	0.98	260M bytes	1.0
16M bytes	0.90	16M bytes	0.99		
64M bytes	0.95	64M bytes	0.995		

- A. What should be the minimum sizes of level 1 and 2 memories to achieve an average access time of less than $100nsec$?
- B. What is the average access time achieved using the chosen sizes of level 1 and level 2 memories?

gate1996 co-and-architecture cache-memory normal

Answer key

1.5.8 Cache Memory: GATE CSE 1998 | Question: 18

For a set-associative Cache organization, the parameters are as follows:

t_c	Cache Access Time
t_m	Main memory access time
l	Number of sets
b	Block size
$k \times b$	Set size

Calculate the hit ratio for a loop executed 100 times where the size of the loop is $n \times b$, and $n = k \times m$ is a non-zero integer and $1 \leq m \leq l$.

Give the value of the hit ratio for $l = 1$.

gate1998 co-and-architecture cache-memory descriptive

Answer key

1.5.9 Cache Memory: GATE CSE 1999 | Question: 1.22

The main memory of a computer has 2 cm blocks while the cache has 2 c blocks. If the cache uses the set associative mapping scheme with 2 blocks per set, then block k of the main memory maps to the set:

- A. $(k \bmod m)$ of the cache
- B. $(k \bmod c)$ of the cache
- C. $(k \bmod 2c)$ of the cache
- D. $(k \bmod 2\text{ cm})$ of the cache

gate1999 co-and-architecture cache-memory normal

Answer key

1.5.10 Cache Memory: GATE CSE 2001 | Question: 1.7, ISRO2008-18

More than one word are put in one cache block to:

- A. exploit the temporal locality of reference in a program
- B. exploit the spatial locality of reference in a program
- C. reduce the miss penalty
- D. none of the above

gatecse-2001 co-and-architecture easy cache-memory isro2008

Answer key

1.5.11 Cache Memory: GATE CSE 2001 | Question: 9

A CPU has 32 — bit memory address and a 256 KB cache memory. The cache is organized as a 4 — way set associative cache with cache block size of 16 bytes.

- A. What is the number of sets in the cache?
- B. What is the size (in bits) of the tag field per cache block?

- C. What is the number and size of comparators required for tag matching?
- D. How many address bits are required to find the byte offset within a cache block?
- E. What is the total amount of extra memory (in bytes) required for the tag bits?

gatecse-2001 co-and-architecture cache-memory normal descriptive

Answer key

1.5.12 Cache Memory: GATE CSE 2002 | Question: 10



In a C program, an array is declared as `float A[2048]`. Each array element is 4 Bytes in size, and the starting address of the array is `0x00000000`. This program is run on a computer that has a direct mapped data cache of size 8 Kbytes, with block (line) size of 16 Bytes.

- A. Which elements of the array conflict with element `A[0]` in the data cache? Justify your answer briefly.
- B. If the program accesses the elements of this array one by one in reverse order i.e., starting with the last element and ending with the first element, how many data cache misses would occur? Justify your answer briefly. Assume that the data cache is initially empty and that no other data or instruction accesses are to be considered.

gatecse-2002 co-and-architecture cache-memory normal descriptive

Answer key

1.5.13 Cache Memory: GATE CSE 2004 | Question: 65



Consider a small two-way set-associative cache memory, consisting of four blocks. For choosing the block to be replaced, use the least recently used (LRU) scheme. The number of cache misses for the following sequence of block addresses is:

8, 12, 0, 12, 8.

- A. 2
- B. 3
- C. 4
- D. 5

gatecse-2004 co-and-architecture cache-memory normal

Answer key

1.5.14 Cache Memory: GATE CSE 2005 | Question: 67



Consider a direct mapped cache of size 32 KB with block size 32 bytes. The CPU generates 32 bit addresses. The number of bits needed for cache indexing and the number of tag bits are respectively,

- A. 10, 17
- B. 10, 22
- C. 15, 17
- D. 5, 17

gatecse-2005 co-and-architecture cache-memory easy

Answer key

1.5.15 Cache Memory: GATE CSE 2006 | Question: 74



Consider two cache organizations. First one is 32 KB 2-way set associative with 32 byte block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1 multiplexer has latency of 0.6 ns while a k -bit comparator has latency of $\frac{k}{10}$ ns. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2 .

The value of h_1 is:

- A. 2.4 ns
- B. 2.3 ns
- C. 1.8 ns
- D. 1.7 ns

gatecse-2006 co-and-architecture cache-memory normal

Answer key

1.5.16 Cache Memory: GATE CSE 2006 | Question: 75



Consider two cache organizations. First one is 32 kB 2-way set associative with 32 byte block size, the second is of same size but direct mapped. The size of an address is 32 bits in both cases. A 2-to-1

multiplexer has latency of $0.6ns$ while a k -bit comparator has latency of $\frac{k}{10}ns$. The hit latency of the set associative organization is h_1 while that of direct mapped is h_2 . The value of h_2 is:

- A. $2.4ns$ B. $2.3ns$ C. $1.8ns$ D. $1.7ns$

gatecse-2006 co-and-architecture cache-memory normal

Answer key

1.5.17 Cache Memory: GATE CSE 2006 | Question: 80



A CPU has a $32KB$ direct mapped cache with 128 byte-block size. Suppose A is two dimensional array of size 512×512 with elements that occupy 8 -bytes each. Consider the following two C code segments, $P1$ and $P2$.

$P1$:

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        x += A[i][j];
    }
}
```

$P2$:

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        x += A[j][i];
    }
}
```

$P1$ and $P2$ are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by $P1$ be M_1 and that for $P2$ be M_2 .

The value of M_1 is:

- A. 0 B. 2048 C. 16384 D. 262144

gatecse-2006 co-and-architecture cache-memory normal

Answer key

1.5.18 Cache Memory: GATE CSE 2006 | Question: 81



A CPU has a $32KB$ direct mapped cache with 128 byte-block size. Suppose A is two dimensional array of size 512×512 with elements that occupy 8 – bytes each. Consider the following two C code segments, $P1$ and $P2$.

$P1$:

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        x += A[i][j];
    }
}
```

$P2$:

```
for (i=0; i<512; i++)
{
    for (j=0; j<512; j++)
    {
        x += A[j][i];
    }
}
```

$P1$ and $P2$ are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers.

x are in registers. Let the number of cache misses experienced by $P1$ be $M1$ and that for $P2$ be $M2$.

The value of the ratio $\frac{M1}{M2}$:

- A. 0 B. $\frac{1}{16}$ C. $\frac{1}{8}$ D. 16

co-and-architecture cache-memory normal gategce-2006

Answer key

1.5.19 Cache Memory: GATE CSE 2007 | Question: 10



Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20-bit address of a word in main memory. The number of bits in the TAG, LINE and WORD fields are respectively:

- A. 9,6,5 B. 7,7,6 C. 7,5,8 D. 9,5,6

gategce-2007 co-and-architecture cache-memory normal

Answer key

1.5.20 Cache Memory: GATE CSE 2007 | Question: 80



Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location $1100H$. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

How many data misses will occur in total?

- A. 48 B. 50 C. 56 D. 59

gategce-2007 co-and-architecture cache-memory normal

Answer key

1.5.21 Cache Memory: GATE CSE 2007 | Question: 81



Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location $1100H$. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

- A. line 4 to line 11 B. line 4 to line 12
C. line 0 to line 7 D. line 0 to line 8

gategce-2007 co-and-architecture cache-memory normal

Answer key

1.5.22 Cache Memory: GATE CSE 2008 | Question: 35



For inclusion to hold between two cache levels L_1 and L_2 in a multi-level cache hierarchy, which of the following are necessary?

- I. L_1 must be write-through cache
- II. L_2 must be a write-through cache
- III. The associativity of L_2 must be greater than that of L_1
- IV. The L_2 cache must be at least as large as the L_1 cache

- A. IV only B. I and IV only C. I, II and IV only D. I, II, III and IV

gategce-2008 co-and-architecture cache-memory normal

Answer key

1.5.23 Cache Memory: GATE CSE 2008 | Question: 71



Consider a machine with a 2-way set associative data cache of size 64Kbytes and block size 16bytes. The cache is managed using 32 bit virtual addresses and the page size is 4Kbytes. A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;
/*Initialize array ARR to 0.0 */
for(i = 0; i < 1024; i++)
    for(j = 0; j < 1024; j++)
        ARR[i][j] = 0.0;
```

The size of double is 8Bytes. Array **ARR** is located in memory starting at the beginning of virtual page 0xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array **ARR**.

The total size of the tags in the cache directory is:

- A. 32Kbits B. 34Kbits C. 64Kbits D. 68Kbits

gatecse-2008 co-and-architecture cache-memory normal

Answer key

1.5.24 Cache Memory: GATE CSE 2008 | Question: 72



Consider a machine with a 2-way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes. A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;
/*Initialize array ARR to 0.0 */
for(i = 0; i < 1024; i++)
    for(j = 0; j < 1024; j++)
        ARR[i][j] = 0.0;
```

The size of double is 8 bytes. Array **ARR** is located in memory starting at the beginning of virtual page 0xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array **ARR**.

Which of the following array elements have the same cache index as **ARR[0][0]**?

- A. **ARR[0][4]** B. **ARR[4][0]** C. **ARR[0][5]** D. **ARR[5][0]**

gatecse-2008 co-and-architecture cache-memory normal

Answer key

1.5.25 Cache Memory: GATE CSE 2008 | Question: 73



Consider a machine with a 2-way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes. A program to be run on this machine begins as follows:

```
double ARR[1024][1024];
int i, j;
/*Initialize array ARR to 0.0 */
for(i = 0; i < 1024; i++)
    for(j = 0; j < 1024; j++)
        ARR[i][j] = 0.0;
```

The size of double is 8 bytes. Array **ARR** is located in memory starting at the beginning of virtual page 0xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array **ARR**.

The cache hit ratio for this initialization loop is:

A. 0%

B. 25%

C. 50%

D. 75%

gatecse-2008 co-and-architecture cache-memory normal

Answer key

1.5.26 Cache Memory: GATE CSE 2009 | Question: 29

Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 blocks and the request for memory blocks are in the following order:

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155.

Which one of the following memory block will NOT be in cache if LRU replacement policy is used?

A. 3

B. 8

C. 129

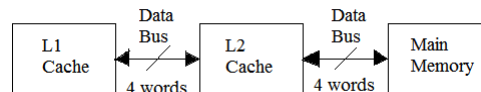
D. 216

gatecse-2009 co-and-architecture cache-memory normal

Answer key

1.5.27 Cache Memory: GATE CSE 2010 | Question: 48

A computer system has an $L1$ cache, an $L2$ cache, and a main memory unit connected as shown below. The block size in $L1$ cache is 4 words. The block size in $L2$ cache is 16 words. The memory access times are 2 nanoseconds, 20 nanoseconds and 200 nanoseconds for $L1$ cache, $L2$ cache and the main memory unit respectively.



When there is a miss in $L1$ cache and a hit in $L2$ cache, a block is transferred from $L2$ cache to $L1$ cache. What is the time taken for this transfer?

A. 2 nanoseconds

B. 20 nanoseconds

C. 22 nanoseconds

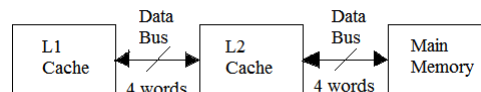
D. 88 nanoseconds

gatecse-2010 co-and-architecture cache-memory normal barc2017

Answer key

1.5.28 Cache Memory: GATE CSE 2010 | Question: 49

A computer system has an $L1$ cache, an $L2$ cache, and a main memory unit connected as shown below. The block size in $L1$ cache is 4 words. The block size in $L2$ cache is 16 words. The memory access times are 2 nanoseconds, 20 nanoseconds and 200 nanoseconds for $L1$ cache, $L2$ cache and the main memory unit respectively.



When there is a miss in both $L1$ cache and $L2$ cache, first a block is transferred from main memory to $L2$ cache, and then a block is transferred from $L2$ cache to $L1$ cache. What is the total time taken for these transfers?

A. 222 nanoseconds

B. 888 nanoseconds

C. 902 nanoseconds

D. 968 nanoseconds

gatecse-2010 co-and-architecture cache-memory normal

Answer key

1.5.29 Cache Memory: GATE CSE 2011 | Question: 43

An 8KB direct-mapped write-back cache is organized as multiple blocks, each size of 32-bytes. The processor generates 32-bit addresses. The cache controller contains the tag information for each cache block comprising of the following.

- 1 valid bit
- 1 modified bit
- As many bits as the minimum needed to identify the memory block mapped in the cache.

What is the total size of memory needed at the cache controller to store meta-data (tags) for the cache?

- A. 4864 bits B. 6144 bits C. 6656 bits D. 5376 bits

gatecse-2011 co-and-architecture cache-memory normal

Answer key

1.5.30 Cache Memory: GATE CSE 2012 | Question: 54



A computer has a 256-KByte, 4-way set associative, write back data cache with block size of 32-Bytes. The processor sends 32-bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

The number of bits in the tag field of an address is

- A. 11 B. 14 C. 16 D. 27

gatecse-2012 co-and-architecture cache-memory normal

Answer key

1.5.31 Cache Memory: GATE CSE 2012 | Question: 55



A computer has a 256-KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

The size of the cache tag directory is:

- A. 160 Kbits B. 136 Kbits C. 40 Kbits D. 32 Kbits

normal gatecse-2012 co-and-architecture cache-memory

Answer key

1.5.32 Cache Memory: GATE CSE 2013 | Question: 20



In a k -way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set $(s + 1)$. The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

- A. $(j \bmod v) * k$ to $(j \bmod v) * k + (k - 1)$
 B. $(j \bmod v)$ to $(j \bmod v) + (k - 1)$
 C. $(j \bmod k)$ to $(j \bmod k) + (v - 1)$
 D. $(j \bmod k) * v$ to $(j \bmod k) * v + (v - 1)$

gatecse-2013 co-and-architecture cache-memory normal

Answer key

1.5.33 Cache Memory: GATE CSE 2014 | Set 1 | Question: 44



An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above by k . What is the miss ratio if the access sequence is passed through a cache of associativity $A \geq k$ exercising least-recently-used replacement policy?

- A. $\left(\frac{n}{N}\right)$ B. $\left(\frac{1}{N}\right)$ C. $\left(\frac{1}{A}\right)$ D. $\left(\frac{k}{n}\right)$

gatecse-2014-set1 co-and-architecture cache-memory normal

Answer key

1.5.34 Cache Memory: GATE CSE 2014 | Set 2 | Question: 43



In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Which one of the following statements is correct in this context?

- A. A smaller block size implies better spatial locality
- B. A smaller block size implies a smaller cache tag and hence lower cache tag overhead
- C. A smaller block size implies a larger cache tag and hence lower cache hit time
- D. A smaller block size incurs a lower cache miss penalty

gatecse-2014-set2 co-and-architecture cache-memory normal

Answer key

1.5.35 Cache Memory: GATE CSE 2014 | Set 2 | Question: 44



If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

- A. Width of tag comparator
- B. Width of set index decoder
- C. Width of way selection multiplexer
- D. Width of processor to main memory data bus

gatecse-2014-set2 co-and-architecture cache-memory normal

Answer key

1.5.36 Cache Memory: GATE CSE 2014 | Set 2 | Question: 9



A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is ____.

gatecse-2014-set2 co-and-architecture cache-memory numerical-answers normal

Answer key

1.5.37 Cache Memory: GATE CSE 2014 | Set 3 | Question: 44



The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is ____.

gatecse-2014-set3 co-and-architecture cache-memory numerical-answers normal

Answer key

1.5.38 Cache Memory: GATE CSE 2015 | Set 2 | Question: 24



Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is ____.

gatecse-2015-set2 co-and-architecture cache-memory easy numerical-answers

Answer key

1.5.39 Cache Memory: GATE CSE 2015 | Set 3 | Question: 14



Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and cache line addresses (in hex) for main memory address $(E201F)_{16}$?

- A. E, 201
- B. F, 201
- C. E, E20
- D. 2, 01F

gatecse-2015-set3 co-and-architecture cache-memory normal

Answer key

1.5.40 Cache Memory: GATE CSE 2016 | Set 2 | Question: 32



The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is _____ bits.

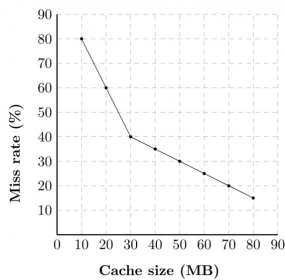
gatecse-2016-set2 co-and-architecture cache-memory normal numerical-answers

Answer key

1.5.41 Cache Memory: GATE CSE 2016 | Set 2 | Question: 50



A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

gatecse-2016-set2 co-and-architecture cache-memory normal numerical-answers

Answer key

1.5.42 Cache Memory: GATE CSE 2017 | Set 1 | Question: 25



Consider a two-level cache hierarchy with L_1 and L_2 caches. An application incurs 1.4 memory accesses per instruction on average. For this application, the miss rate of L_1 cache is 0.1; the L_2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L_2 expressed correct to two decimal places is _____.

gatecse-2017-set1 co-and-architecture cache-memory numerical-answers

Answer key

1.5.43 Cache Memory: GATE CSE 2017 | Set 1 | Question: 54



A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____ bits.

gatecse-2017-set1 co-and-architecture cache-memory normal numerical-answers

Answer key

1.5.44 Cache Memory: GATE CSE 2017 | Set 2 | Question: 29



In a two-level cache system, the access times of L_1 and L_2 caches are 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2 . The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are

- A. 0.111 and 0.056 B. 0.056 and 0.111 C. 0.0892 and 0.1784 D. 0.1784 and 0.0892

gatecse-2017-set2 cache-memory co-and-architecture normal

Answer key

1.5.45 Cache Memory: GATE CSE 2017 | Set 2 | Question: 45



The read access times and the hit ratios for different caches in a memory hierarchy are as given below:

Cache	Read access time (in nanoseconds)	Hit ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the write-back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____

gatecse-2017-set2 co-and-architecture cache-memory numerical-answers

Answer key

1.5.46 Cache Memory: GATE CSE 2017 | Set 2 | Question: 53



Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____

gatecse-2017-set2 co-and-architecture cache-memory numerical-answers

Answer key

1.5.47 Cache Memory: GATE CSE 2018 | Question: 34



The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K -way set-associative cache memory, the length (in number of bits) of the tag field is

- A. $P - N - \log_2 K$ B. $P - N + \log_2 K$
 C. $P - N - M - W - \log_2 K$ D. $P - N - M - W + \log_2 K$

gatecse-2018 co-and-architecture cache-memory normal two-marks

Answer key

1.5.48 Cache Memory: GATE CSE 2019 | Question: 1



A certain processor uses a fully associative cache of size 16 kB, The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the Tag and the Index fields respectively in the addresses generated by the processor?

- A. 24 bits and 0 bits B. 28 bits and 4 bits
 C. 24 bits and 4 bits D. 28 bits and 0 bits

gatecse-2019 co-and-architecture cache-memory normal one-mark

Answer key

1.5.49 Cache Memory: GATE CSE 2019 | Question: 45



A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ $\times 10^6$ bytes/sec.

gatecse-2019 numerical-answers co-and-architecture cache-memory two-marks

Answer key

1.5.50 Cache Memory: GATE CSE 2020 | Question: 21



A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____.

gatecse-2020 numerical-answers co-and-architecture cache-memory one-mark

Answer key

1.5.51 Cache Memory: GATE CSE 2020 | Question: 30



A computer system with a word length of 32 bits has a 16 MB byte- addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.

- $A1 = 0x42C8A4$,
- $A2 = 0x546888$,
- $A3 = 0x6A289C$,
- $A4 = 0x5E4880$

Which one of the following is TRUE?

- A. $A1$ and $A4$ are mapped to different cache sets.
- B. $A2$ and $A3$ are mapped to the same cache set.
- C. $A3$ and $A4$ are mapped to the same cache set.
- D. $A1$ and $A3$ are mapped to the same cache set.

gatecse-2020 co-and-architecture cache-memory two-marks

Answer key

1.5.52 Cache Memory: GATE CSE 2021 | Set 1 | Question: 22



Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32 KB ($1 \text{ KB} = 2^{10}$ bytes), and each cache block is of size 64 bytes.

The size of the tag field is _____ bits.

gatecse-2021-set1 co-and-architecture cache-memory numerical-answers one-mark

Answer key

1.5.53 Cache Memory: GATE CSE 2021 | Set 2 | Question: 19



Consider a set-associative cache of size 2KB ($1 \text{ KB} = 2^{10}$ bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32 -bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____

gatecse-2021-set2 numerical-answers co-and-architecture cache-memory one-mark

Answer key

1.5.54 Cache Memory: GATE CSE 2021 | Set 2 | Question: 27



Assume a two-level inclusive cache hierarchy, $L1$ and $L2$, where $L2$ is the larger of the two. Consider the following statements.

- S_1 : Read misses in a write through $L1$ cache do not result in writebacks of dirty lines to the $L2$
- S_2 : Write allocate policy *must* be used in conjunction with write through caches and no-write allocate policy is used with writeback caches.

Which of the following statements is correct?

- A. S_1 is true and S_2 is false
- B. S_1 is false and S_2 is true
- C. S_1 is true and S_2 is true
- D. S_1 is false and S_2 is false

gatecse-2021-set2 co-and-architecture cache-memory two-marks

Answer key

1.5.55 Cache Memory: GATE CSE 2022 | Question: 14



Let **WB** and **WT** be two set associative cache organizations that use **LRU** algorithm for cache block replacement. **WB** is a write back cache and **WT** is a write through cache. Which of the following statements is/are **FALSE**?

- A. Each cache block in **WB** and **WT** has a dirty bit.
- B. Every write hit in **WB** leads to a data transfer from cache to main memory.
- C. Eviction of a block from **WT** will not lead to data transfer from cache to main memory.
- D. A read miss in **WB** will never lead to eviction of a dirty block from **WB**.

gatecse-2022 co-and-architecture cache-memory multiple-selects one-mark

Answer key

1.5.56 Cache Memory: GATE CSE 2022 | Question: 23



A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (*rounded off to two decimal places*) needed after the optimization such that it should not increase the average memory access time is _____.

gatecse-2022 numerical-answers co-and-architecture cache-memory one-mark

Answer key

1.5.57 Cache Memory: GATE CSE 2023 | Question: 54



An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into **TAG**, **INDEX**, and **BLOCK OFFSET**.

The number of bits in the **TAG** is _____.

gatecse-2023 co-and-architecture cache-memory numerical-answers two-marks

Answer key

1.5.58 Cache Memory: GATE CSE 2024 | Set 1 | Question: 43



Consider two set-associative cache memory architectures: **WBC**, which uses the write back policy, and **WTC**, which uses the write through policy. Both of them use the **LRU** (*Least Recently Used*) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are **TRUE**?

- A. A read miss in **WBC** never evicts a dirty block
- B. A read miss in **WTC** never triggers a write back operation of a cache block to main memory
- C. A write hit in **WBC** can modify the value of the dirty bit of a cache block
- D. A write miss in **WTC** always writes the victim cache block to main memory before loading the missed block to the cache

gatecse-2024-set1 co-and-architecture cache-memory multiple-selects two-marks

Answer key

1.5.59 Cache Memory: GATE CSE 2024 | Set 1 | Question: 46

A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (*rounded off to two decimal places*) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____.

gatecse-2024-set1 numerical-answers co-and-architecture cache-memory two-marks

Answer key

1.5.60 Cache Memory: GATE CSE 2026 | Set 1 | Question: 28

The size of the physical address space of a processor is 2^{32} bytes. The capacity of a cache memory unit is 2^{23} bytes. The cache block size is 128 bytes. The cache memory unit can be built as a direct mapped cache or as a K -way set-associative cache, where $K = 2^L$ and $L \in \{1, 2, 3\}$. Let the length of the TAG field be M bits for the direct mapped cache, and N bits for the set-associative cache.

Which one of the following options is true?

- A. $N = M + L$ B. $N = M - L$ C. $N = M + K$ D. $N = M - K$

gatecse-2026-set1 co-and-architecture two-marks cache-memory

Answer key

1.5.61 Cache Memory: GATE CSE 2026 | Set 1 | Question: 44

Consider a system that has a cache memory unit and a memory management unit (MMU). The address input to the cache memory is a physical address. The MMU has a translation lookaside buffer (TLB). Assume that when a page is evicted from the main memory, the corresponding blocks in the cache are marked as invalid.

For a given memory reference, which of the following sequences of events can NEVER happen?

- A. TLB miss, Page table hit, Cache hit
 B. TLB hit, Page table miss, Cache hit
 C. TLB miss, Page table miss, Cache hit
 D. TLB miss, Page table miss, Cache miss

gatecse-2026-set1 co-and-architecture cache-memory two-marks multiple-selects

Answer key

1.5.62 Cache Memory: GATE IT 2004 | Question: 12, ISRO2016-77

Consider a system with 2 level cache. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10 ns, and 500 ns respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

- A. 13.0 B. 12.8 C. 12.6 D. 12.4

gateit-2004 co-and-architecture cache-memory normal isro2016

Answer key

1.5.63 Cache Memory: GATE IT 2004 | Question: 48

Consider a fully associative cache with 8 cache blocks (numbered 0 – 7) and the following sequence of memory block requests:

4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7

If LRU replacement policy is used, which cache block will have memory block 7?

- A. 4 B. 5 C. 6 D. 7

gateit-2004 co-and-architecture cache-memory normal

Answer key

1.5.64 Cache Memory: GATE IT 2005 | Question: 61



Consider a 2-way set associative cache memory with 4 sets and total 8 cache blocks (0 – 7) and a main memory with 128 blocks (0 – 127). What memory blocks will be present in the cache after the following sequence of memory block references if **LRU** policy is used for cache block replacement. Assuming that initially the cache did not have any memory block from the current job?

0 5 3 9 7 0 16 55

A. 0 3 5 7 16 55

B. 0 3 5 7 9 16 55

C. 0 5 7 9 16 55

D. 3 5 7 9 16 55

gateit-2005 co-and-architecture cache-memory normal

Answer key

1.5.65 Cache Memory: GATE IT 2006 | Question: 42



A cache line is 64 bytes. The main memory has latency 32 ns and bandwidth 1 GBytes/s. The time required to fetch the entire cache line from the main memory is:

A. 32 ns

B. 64 ns

C. 96 ns

D. 128 ns

gateit-2006 co-and-architecture cache-memory normal

Answer key

1.5.66 Cache Memory: GATE IT 2006 | Question: 43



A computer system has a level-1 instruction cache (*I*-cache), a level-1 data cache (*D*-cache) and a level-2 cache (*L2*-cache) with the following specifications:

	Capacity	Mapping Method	Block Size
<i>I</i> -Cache	4K words	Direct mapping	4 words
<i>D</i> -Cache	4K words	2 -way set associative mapping	4 words
<i>L2</i> -Cache	64K words	4-way set associative mapping	16 words

The length of the physical address of a word in the main memory is 30 bits. The capacity of the tag memory in the *I*-cache, *D*-cache and *L2*-cache is, respectively,

A. 1 K x 18-bit, 1 K x 19-bit, 4 K x 16-bit

B. 1 K x 16-bit, 1 K x 19-bit, 4 K x 18-bit

C. 1 K x 16-bit, 512 x 18-bit, 1 K x 16-bit

D. 1 K x 18-bit, 512 x 18-bit, 1 K x 18-bit

gateit-2006 co-and-architecture cache-memory normal

Answer key

1.5.67 Cache Memory: GATE IT 2007 | Question: 37



Consider a Direct Mapped Cache with 8 cache blocks (numbered 0 – 7). If the memory block requests are in the following order

3, 5, 2, 8, 0, 63, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24.

Which of the following memory blocks will not be in the cache at the end of the sequence ?

A. 3

B. 18

C. 20

D. 30

gateit-2007 co-and-architecture cache-memory normal

Answer key

1.5.68 Cache Memory: GATE IT 2008 | Question: 80



Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB.

The number of bits in the TAG, SET and WORD fields, respectively are:

A. 7,6,7

B. 8,5,7

C. 8,6,6

D. 9,4,7

gateit-2008 co-and-architecture cache-memory normal

Answer key

1.5.69 Cache Memory: GATE IT 2008 | Question: 81



Consider a computer with a 4-ways set-associative mapped cache of the following characteristics: a total of 1 MB of main memory, a word size of 1 byte, a block size of 128 words and a cache size of 8 KB.

While accessing the memory location 0C795H by the CPU, the contents of the TAG field of the corresponding cache line is:

A. 000011000

B. 110001111

C. 00011000

D. 110010101

gateit-2008 co-and-architecture cache-memory normal

Answer key

1.6

Conflict Misses (1)

1.6.1 Conflict Misses: GATE CSE 2017 | Set 1 | Question: 51



Consider a 2-way set associative cache with 256 blocks and uses LRU replacement. Initially the cache is empty. Conflict misses are those misses which occur due to the contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of access to memory blocks :

$\{0, 128, 256, 128, 0, 128, 256, 128, 1, 129, 257, 129, 1, 129, 257, 129\}$

is repeated 10 times. The number of *conflict misses* experienced by the cache is _____ .

gatecse-2017-set1 co-and-architecture cache-memory conflict-misses normal numerical-answers

Answer key

1.7

Control Unit (1)

1.7.1 Control Unit: GATE CSE 1987 | Question: 1-vi



A microprogrammed control unit

- A. Is faster than a hard-wired control unit.
- B. Facilitates easy implementation of new instruction.
- C. Is useful when very small programs are to be run.
- D. Usually refers to the control unit of a microprocessor.

gate1987 co-and-architecture control-unit microprogramming

Answer key

1.8

DMA (8)

Practice Test: Test 1 (12Q)

1.8.1 DMA: GATE CSE 2004 | Question: 68



A hard disk with a transfer rate of 10 Mbytes/second is constantly transferring data to memory using DMA. The processor runs at 600 MHz, and takes 300 and 900 clock cycles to initiate and complete DMA transfer respectively. If the size of the transfer is 20 Kbytes, what is the percentage of processor time consumed for the transfer operation?

A. 5.0%

B. 1.0%

C. 0.5%

D. 0.1%

gatecse-2004 dma normal co-and-architecture

Answer key

1.8.2 DMA: GATE CSE 2005 | Question: 70



Consider a disk drive with the following specifications:

16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, rotation speed 3000 rpm. The disk is operated in cycle stealing mode whereby whenever one 4 byte word is ready it is sent to memory; similarly, for writing, the disk interface reads a 4 byte word from the memory in each DMA cycle. Memory cycle time is 40 nsec. The maximum percentage of time that the CPU gets blocked during DMA operation is:

- A. 10 B. 25 C. 40 D. 50

gatecse-2005 co-and-architecture disk normal dma

Answer key

1.8.3 DMA: GATE CSE 2016 | Set 1 | Question: 31



The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _____.

gatecse-2016-set1 co-and-architecture dma normal numerical-answers

Answer key

1.8.4 DMA: GATE CSE 2021 | Set 2 | Question: 20



Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through *cycle stealing* at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second.

gatecse-2021-set2 numerical-answers co-and-architecture dma one-mark

Answer key

1.8.5 DMA: GATE CSE 2022 | Question: 7



Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?

- A. DMA based I/O transfer B. Interrupt driven I/O transfer
C. Polling based I/O transfer D. Programmed I/O transfer

gatecse-2022 co-and-architecture dma one-mark

Answer key

1.8.6 DMA: GATE CSE 2024 | Set 1 | Question: 5



Which one of the following statements is FALSE?

- A. In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle
B. For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode
C. Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism
D. The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

gatecse-2024-set1 co-and-architecture dma one-mark

Answer key

1.8.7 DMA: GATE CSE 2024 | Set 2 | Question: 1



Consider a computer with a 4MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (*in bits per second*) of the DMA controller if 1% of the processor cycles are used for DMA?

A. 2,56,000

B. 3,200

C. 25,60,000

D. 32,000

gatecse-2024-set2 co-and-architecture dma one-mark

Answer key

1.8.8 DMA: GATE IT 2004 | Question: 51



The storage area of a disk has the innermost diameter of 10 cm and outermost diameter of 20 cm. The maximum storage density of the disk is 1400 bits/cm. The disk rotates at a speed of 4200 RPM. The main memory of a computer has 64-bit word length and 1 μ s cycle time. If cycle stealing is used for data transfer from the disk, the percentage of memory cycles stolen for transferring one word is

A. 0.5%

B. 1%

C. 5%

D. 10%

gateit-2004 co-and-architecture dma normal

Answer key

1.9 DRAM (1)

1.9.1 DRAM: GATE CSE 1919 | Question: 2



The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A_{15} to A_0 . What is the range of address (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



A. C800 to CFFF

B. CA00 to CAFF

C. C800 to C8FF

D. DA00 to DFFF

gatecse-2019 co-and-architecture dram memory-interfacing one-mark

Answer key

1.10 Data Dependency (4)

1.10.1 Data Dependency: GATE CSE 2005 | Question: 68



A 5 stage pipelined CPU has the following sequence of stages:

- IF – instruction fetch from instruction memory
- RD – Instruction decode and register read
- EX – Execute: ALU operation for data and address computation
- MA – Data memory access – for write access, the register read at RD state is used.
- WB – Register write back

Consider the following sequence of instructions:

- $I_1: L\ R0, loc\ 1; R0 \leftarrow M[loc1]$
- $I_2: A\ R0, R0; R0 \leftarrow R0 + R0$
- $I_3: S\ R2, R0; R2 \leftarrow R2 - R0$

Let each stage take one clock cycle.

What is the number of clock cycles taken to complete the above sequence of instructions starting from the fetch of I_1 ?

A. 8

B. 10

C. 12

D. 15

gatecse-2005 co-and-architecture pipelining normal data-dependency

Answer key

1.10.2 Data Dependency: GATE CSE 2008 | Question: 36



Which of the following are NOT true in a pipelined processor?

- I. Bypassing can handle all RAW hazards
- II. Register renaming can eliminate all register carried WAR hazards
- III. Control hazard penalties can be eliminated by dynamic branch prediction

A. I and II only B. I and III only C. II and III only D. I, II and III

gatecse-2008 pipelining co-and-architecture normal data-dependency

Answer key

1.10.3 Data Dependency: GATE CSE 2015 | Set 3 | Question: 47



Consider the following code sequence having five instructions from I_1 to I_5 . Each of these instructions has the following format.

OP Ri, Rj, Rk

Where operation OP is performed on contents of registers Rj and Rk and the result is stored in register Ri.

I_1 : ADD R1, R2, R3

I_2 : MUL R7, R1, R3

I_3 : SUB R4, R1, R5

I_4 : ADD R3, R2, R4

I_5 : MUL R7, R8, R9

Consider the following three statements.

S1: There is an anti-dependence between instructions I_2 and I_5

S2: There is an anti-dependence between instructions I_2 and I_4

S3: Within an instruction pipeline an anti-dependence always creates one or more stalls

Which one of the above statements is/are correct?

- A. Only S1 is true B. Only S2 is true
C. Only S1 and S3 are true D. Only S2 and S3 are true

gatecse-2015-set3 co-and-architecture pipelining data-dependency normal

Answer key

1.10.4 Data Dependency: GATE IT 2007 | Question: 39



Data forwarding techniques can be used to speed up the operation in presence of data dependencies. Consider the following replacements of LHS with RHS.

- i. $R1 \rightarrow Loc, Loc \rightarrow R2 \equiv R1 \rightarrow R2, R1 \rightarrow Loc$
- ii. $R1 \rightarrow Loc, Loc \rightarrow R2 \equiv R1 \rightarrow R2$
- iii. $R1 \rightarrow Loc, R2 \rightarrow Loc \equiv R1 \rightarrow Loc$
- iv. $R1 \rightarrow Loc, R2 \rightarrow Loc \equiv R2 \rightarrow Loc$

In which of the following options, will the result of executing the RHS be the same as executing the LHS irrespective of the instructions that follow ?

- A. i and iii B. i and iv
C. ii and iii D. ii and iv

gateit-2007 data-dependency co-and-architecture

Answer key

1.11 Data Hazards (1)

1.11.1 Data Hazards: GATE CSE 2026 | Set 1 | Question: 6



Which one of the following dependencies among the register operands of different instructions can cause a data hazard in a pipelined processor?

- A. Read-after-read
C. Write-after-read

- B. Read-after-write
D. Write-after-write

gatecse-2026-set1 co-and-architecture data-hazards easy one-mark

Answer key

1.12

Data Path (7)

📌 Practice Test: Test 1 (6Q)

1.12.1 Data Path: GATE CSE 1990 | Question: 8a



A single bus CPU consists of four general purpose register, namely, $R0, \dots, R3$, ALU, MAR, MDR, PC, SP and IR (Instruction Register). Assuming suitable microinstructions, write a microroutine for the instruction, **ADD $R0, R1$** .

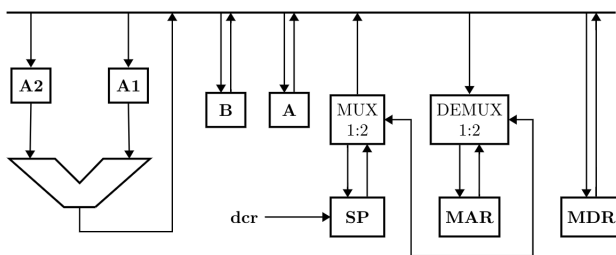
gate1990 descriptive co-and-architecture data-path

Answer key

1.12.2 Data Path: GATE CSE 2001 | Question: 2.13



Consider the following data path of a simple non-pipelined CPU. The registers A, B, A_1, A_2 , MDR, the bus and the ALU are 8-bit wide. SP and MAR are 16-bit registers. The MUX is of size $8 \times (2:1)$ and the DEMUX is of size $8 \times (1:2)$. Each memory operation takes 2 CPU clock cycles and uses MAR (Memory Address Register) and MDR (Memory Date Register). SP can be decremented locally.



The CPU instruction "push r " where, $r = A$ or B has the specification

- $M[SP] \leftarrow r$
- $SP \leftarrow SP - 1$

How many CPU clock cycles are required to execute the "push r " instruction?

- A. 2 B. 3 C. 4 D. 5

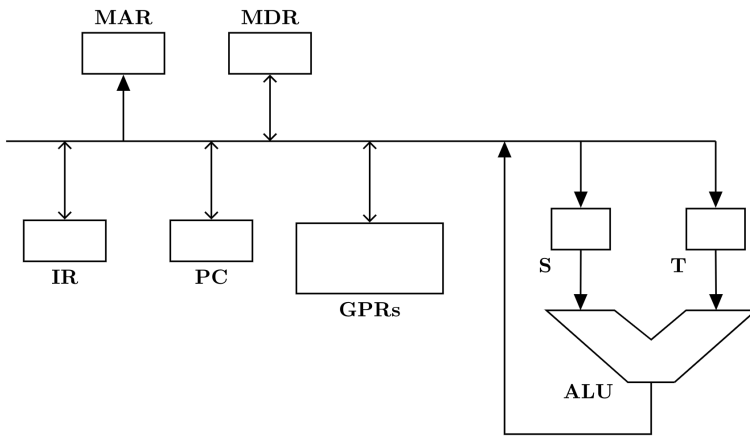
gatecse-2001 co-and-architecture data-path machine-instruction normal

Answer key

1.12.3 Data Path: GATE CSE 2005 | Question: 79



Consider the following data path of a CPU.



The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation – the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR.

The instruction “add R0, R1” has the register transfer interpretation $R0 \leftarrow R0 + R1$. The minimum number of clock cycles needed for execution cycle of this instruction is:

- A. 2 B. 3 C. 4 D. 5

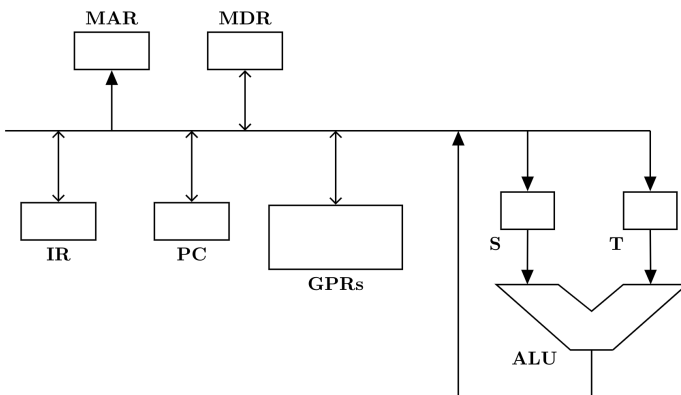
gatecse-2005 co-and-architecture machine-instruction data-path normal

Answer key

1.12.4 Data Path: GATE CSE 2005 | Question: 80



Consider the following data path of a CPU.



The ALU, the bus and all the registers in the data path are of identical size. All operations including incrementation of the PC and the GPRs are to be carried out in the ALU. Two clock cycles are needed for memory read operation – the first one for loading address in the MAR and the next one for loading data from the memory bus into the MDR.

The instruction “call Rn, sub” is a two word instruction. Assuming that PC is incremented during the fetch cycle of the first word of the instruction, its register transfer interpretation is

$$R_n \leftarrow PC + 1;$$

$$PC \leftarrow M[PC];$$

The minimum number of CPU clock cycles needed during the execution cycle of this instruction is:

- A. 2 B. 3 C. 4 D. 5

co-and-architecture normal gatecse-2005 data-path machine-instruction

Answer key

1.12.5 Data Path: GATE CSE 2016 | Set 2 | Question: 30



Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G , respectively. Given two instances of U_F and two instances of U_G , it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

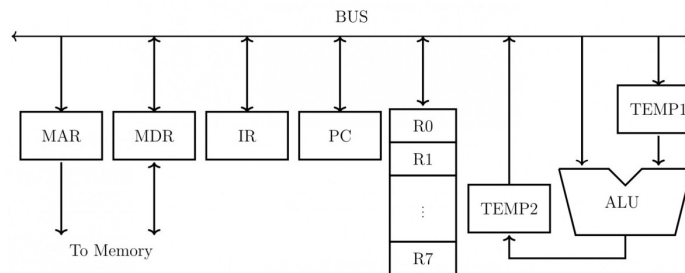
gatecse-2016-set2 co-and-architecture data-path normal numerical-answers

Answer key

1.12.6 Data Path: GATE CSE 2020 | Question: 4



Consider the following data path diagram.



Consider an instruction: $R0 \leftarrow R1 + R2$. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r$, $TEMP1_r$, ALU_{add} , $TEMP2_w$
2. $R1_r$, $TEMP1_w$
3. PC_r , MAR_w , MEM_r
4. $TEMP2_r$, $R0_w$
5. MDR_r , IR_w

Which one of the following is the correct order of execution of the above steps?

- A. 2,1,4,5,3 B. 1,2,4,3,5 C. 3,5,2,1,4 D. 3,5,1,2,4

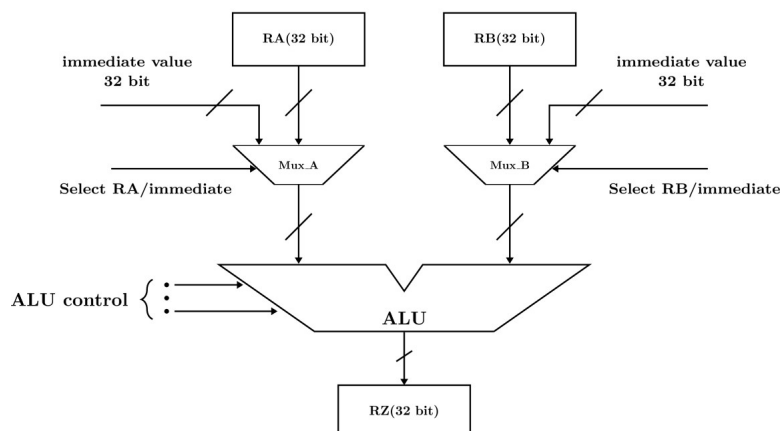
gatecse-2020 co-and-architecture data-path one-mark

Answer key

1.12.7 Data Path: GATE CSE 2025 | Set 1 | Question: 17



A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?



- A. The data path can implement arithmetic operations involving two registers.
 B. The data path can implement arithmetic operations involving one register and one immediate value.

- C. The data path can implement arithmetic operations involving two immediate values.
 D. The data path can only implement arithmetic operations involving one register and one immediate value.

gatecse2025-set1 co-and-architecture data-path multiple-selects one-mark

Answer key

1.13 Direct Mapping (5)

1.13.1 Direct Mapping: GATE CSE 2022 | Question: 44



Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

- A. Every access to S is a hit.
 B. Once P is brought to the cache it is never evicted.
 C. At the end of the execution only R and S reside in the cache.
 D. Every access to R evicts Q from the cache.

gatecse-2022 co-and-architecture direct-mapping multiple-selects two-marks cache-memory

Answer key

1.13.2 Direct Mapping: GATE CSE 2025 | Set 1 | Question: 26



Consider a memory system with 1M bytes of main memory and 16 K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the tag values? [Assume memory is byte addressable, $1\text{ K} = 2^{10}$, $1\text{ M} = 2^{20}$.]

- A. 6×2^{10} B. 8×2^{10} C. 2^{12} D. 2^{14}

gatecse2025-set1 co-and-architecture cache-memory direct-mapping easy two-marks

Answer key

1.13.3 Direct Mapping: GATE CSE 2025 | Set 2 | Question: 29



For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block.

Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?

- A. 64 KB and 4 KB B. 128 KB and 16 KB C. 64 KB and 8 KB D. 128 KB and 6 KB

gatecse2025-set2 co-and-architecture direct-mapping cache-memory two-marks

Answer key

1.13.4 Direct Mapping: GATE CSE 2026 | Set 2 | Question: 42



Consider a system with a processor and a 4 KB direct mapped cache with block size of 16 bytes. The system has a 16 MB physical memory. Four words P, Q, R, and S are accessed by the processor in the same order 10 times. That is, there are a total of 40 memory references in the sequence P, Q, R, S, P, Q, R, S, ...

Assume that the cache memory is initially empty. The physical addresses of the words are given below (1 word = 1 byte).

P: 0x845B32, Q: 0x845B26, R: 0x845B36, S: 0x846B32

Which of the following statements is/are true?

Note: $1\text{ K} = 2^{10}$ and $1\text{ M} = 2^{20}$

- A. Every access to P results in a cache miss
- B. Every access to R results in a cache hit
- C. Every access to Q results in a cache miss
- D. Except the first access to S, all subsequent accesses to S result in cache hits

gatecse-2026-set2 co-and-architecture direct-mapping cache-memory multiple-selects two-marks

Answer key

1.13.5 Direct Mapping: GATE CSE 2026 | Set 2 | Question: 46



Consider a system with 1 MB physical memory and a word length of 1 byte. The system uses a direct mapped cache, with block numbers starting from 0. The word with physical address 0xA2C28 is mapped to the cache block number 176_{10} . The maximum possible size of the cache (in KB) for this configuration is _____. (answer in integer)

Note: $1\text{ K} = 2^{10}$ and $1\text{ M} = 2^{20}$

gatecse-2026-set2 co-and-architecture direct-mapping cache-memory numerical-answers two-marks

Answer key

1.14

Disk (1)

1.14.1 Disk: GATE CSE 2026 | Set 1 | Question: 49



Consider a hard disk with a rotational speed of 15000 rpm. The time to move the read/write head from a track to its adjacent track is 1 millisecond. Initially, the head is on track 0. The number of sectors per track is 400. The sector size is 1024 bytes. It is necessary to transfer data from 10 randomly located sectors in each of the following tracks in the order: 5, 12 and 7.

The total time for the data transfer (in milliseconds) from the hard disk is _____. (rounded off to one decimal place)

gatecse-2026-set1 numerical-answers operating-system two-marks disk

Answer key

1.15

Hazards (1)

1.15.1 Hazards: GATE CSE 2026 | Set 1 | Question: 50



The EX stage of a pipelined processor performs the memory read operations for LOAD instructions, and the operations for the arithmetic and logic instructions. Let t_{EX} denote the time taken by the EX stage to perform the operation for an instruction. For each instruction type, the values of t_{EX} and M (the number of instructions of that type in a sequence of 100 instructions for a program P), are given in the table below.

The duration of the pipeline clock cycle is 1 nanosecond. Assume that the latch time for the interstage buffers in the pipeline is negligible.

Instruction	t_{EX} in nanoseconds	M
LOAD	1.8	15
IMUL	1.5	10
IDIV	2.5	5
FADD	1.7	10
FSUB	1.7	5
FMUL	2.8	15
FDIV	3.2	5
All other instructions	Less than 1.0	35

When program P is executed, the number of clock cycles for which the pipeline is stalled due to structural hazards in the EX stage is _____. (answer in integer)

gatecse-2026-set1 numerical-answers co-and-architecture pipelining hazards two-marks

Answer key

1.16 IO Handling (8)

1.16.1 IO Handling: GATE CSE 1987 | Question: 2a



State whether the following statements are TRUE or FALSE

In a microprocessor-based system, if a bus (DMA) request and an interrupt request arrive simultaneously, the microprocessor attends first to the bus request.

gate1987 co-and-architecture interrupts io-handling true-false

Answer key

1.16.2 IO Handling: GATE CSE 1987 | Question: 2b



State whether the following statements are TRUE or FALSE:

Data transfer between a microprocessor and an I/O device is usually faster in memory-mapped-I/O scheme than in I/O-mapped -I/O scheme.

gate1987 co-and-architecture io-handling true-false

Answer key

1.16.3 IO Handling: GATE CSE 1990 | Question: 4-ii



State whether the following statements are TRUE or FALSE with reason:

The data transfer between memory and I/O devices using programmed I/O is faster than interrupt-driven I/O.

gate1990 true-false co-and-architecture io-handling interrupts

Answer key

1.16.4 IO Handling: GATE CSE 1996 | Question: 1.24



For the daisy chain scheme of connecting I/O devices, which of the following statements is true?

- A. It gives non-uniform priority to various devices
- B. It gives uniform priority to all devices
- C. It is only useful for connecting slow devices to a processor device
- D. It requires a separate interrupt pin on the processor for each device

Answer key

1.16.5 IO Handling: GATE CSE 1996 | Question: 25



A hard disk is connected to a 50 MHz processor through a DMA controller. Assume that the initial set-up of a DMA transfer takes 1000 clock cycles for the processor, and assume that the handling of the interrupt at DMA completion requires 500 clock cycles for the processor. The hard disk has a transfer rate of 2000 Kbytes/sec and average block transferred is 4 K bytes. What fraction of the processor time is consumed by the disk, if the disk is actively transferring 100% of the time?

Answer key

1.16.6 IO Handling: GATE CSE 1997 | Question: 2.4



The correct matching for the following pairs is:

- | | |
|-----------------------------|--------------------|
| (A) DMA I/O | (1) High speed RAM |
| (B) Cache | (2) Disk |
| (C) Interrupt I/O | (3) Printer |
| (D) Condition Code Register | (4) ALU |

A. A – 4 B – 3 C – 1 D – 2

B. A – 2 B – 1 C – 3 D – 4

C. A – 4 B – 3 C – 2 D – 1

D. A – 2 B – 3 C – 4 D – 1

Answer key

1.16.7 IO Handling: GATE CSE 2008 | Question: 64, ISRO2009-13



Which of the following statements about synchronous and asynchronous I/O is NOT true?

- An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O
- In both synchronous and asynchronous I/O, an ISR (Interrupt Service Routine) is invoked after completion of the I/O
- A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O
- In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

Answer key

1.16.8 IO Handling: GATE CSE 2011 | Question: 28



On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```

Initialize the address register
Initialize the count to 500
LOOP: Load a byte from device
Store in memory at address given by address register
Increment the address register
Decrement the count
If count != 0 go to LOOP
  
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

- A. 3.4 B. 4.4 C. 5.1 D. 6.7

gatecse-2011 co-and-architecture dma normal io-handling

Answer key

1.17 Instruction Execution (7)

 **Practice Test:** Test 1 (7Q)

1.17.1 Instruction Execution: GATE CSE 1990 | Question: 4-iii



State whether the following statements are TRUE or FALSE with reason:

The flags are affected when conditional CALL or JUMP instructions are executed.

gate1990 true-false co-and-architecture instruction-execution

Answer key

1.17.2 Instruction Execution: GATE CSE 1992 | Question: 01-iv



Many of the advanced microprocessors prefetch instructions and store it in an instruction buffer to speed up processing. This speed up is achieved because _____

gate1992 co-and-architecture easy instruction-execution fill-in-the-blanks

Answer key

1.17.3 Instruction Execution: GATE CSE 1995 | Question: 1.2



Which of the following statements is true?

- A. ROM is a Read/Write memory B. PC points to the last instruction that was executed
C. Stack works on the principle of LIFO D. All instructions affect the flags

gate1995 co-and-architecture normal instruction-execution

Answer key

1.17.4 Instruction Execution: GATE CSE 2002 | Question: 1.13



Which of the following is not a form of memory

- A. instruction cache B. instruction register
C. instruction opcode D. translation look-a-side buffer

gatecse-2002 co-and-architecture easy instruction-execution

Answer key

1.17.5 Instruction Execution: GATE CSE 2006 | Question: 43



Consider a new instruction named branch-on-bit-set (mnemonic bbs). The instruction “bbs reg, pos, label” jumps to label if bit in position pos of register operand reg is one. A register is 32 -bits wide and the bits are numbered 0 to 31, bit in position 0 being the least significant. Consider the following emulation of this instruction on a processor that does not have bbs implemented.

$temp \leftarrow reg \& mask$

Branch to label if temp is non-zero. The variable temp is a temporary register. For correct emulation, the variable mask must be generated by

- A. $mask \leftarrow 0x1 \ll pos$ B. $mask \leftarrow 0xffffffff \ll pos$
C. $mask \leftarrow pos$ D. $mask \leftarrow 0xf$

gatecse-2006 co-and-architecture normal instruction-execution

Answer key

1.17.6 Instruction Execution: GATE CSE 2017 | Set 1 | Question: 49



Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence

Instr. No.	Instruction
i:	add R2, R3, R4
i+1:	sub R5, R6, R7
i+2:	cmp R1, R9, R10
i+3:	beq R1, Offset

If the target of the branch instruction is i , then the decimal value of the Offset is _____.

gatecse-2017-set1 co-and-architecture normal numerical-answers instruction-execution

Answer key

1.17.7 Instruction Execution: GATE CSE 2025 | Set 2 | Question: 51



An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____. (rounded off to one decimal place)

Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed
Branch	2	2.25×10^8
Load	5	1.20×10^8
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

gatecse2025-set2 co-and-architecture instruction-execution clock-cycles numerical-answers two-marks

Answer key

1.18 Instruction Format (11)

Practice Test: Test 1 (10Q)

1.18.1 Instruction Format: GATE CSE 1988 | Question: 2-ii



Using an expanding opcode encoding for instructions, is it possible to encode all of the following in an instruction format shown in the below figure. Justify your answer.

- 14 double address instructions
- 127 single address instructions
- 60 no address (zero address) instructions

← 4 bits →	← 6 bits →	← 6 bits →
Opcode	Operand 1 Address	Operand 2 Address

gate1988 normal co-and-architecture instruction-format descriptive

Answer key

1.18.2 Instruction Format: GATE CSE 1992 | Question: 01-vi



In an 11-bit computer instruction format, the size of address field is 4-bits. The computer uses expanding OP code technique and has 5 two-address instructions and 32 one-address instructions. The number of zero-address instructions it can support is _____

gate1992 co-and-architecture machine-instruction instruction-format normal numerical-answers

Answer key

1.18.3 Instruction Format: GATE CSE 1994 | Question: 3.2



State True or False with one line explanation

Expanding opcode instruction formats are commonly employed in RISC. (Reduced Instruction Set Computers) machines.

gate1994 co-and-architecture machine-instruction instruction-format normal true-false

Answer key

1.18.4 Instruction Format: GATE CSE 2014 | Set 1 | Question: 9



A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is _____

gatecse-2014-set1 co-and-architecture machine-instruction instruction-format numerical-answers normal

Answer key

1.18.5 Instruction Format: GATE CSE 2016 | Set 2 | Question: 31



Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____.

gatecse-2016-set2 instruction-format machine-instruction co-and-architecture normal numerical-answers

Answer key

1.18.6 Instruction Format: GATE CSE 2018 | Question: 51



A processor has 16 integer registers ($R_0, R_1, \dots, R_{15}$) and 64 floating point registers ($F_0, F_1, \dots, F_{63}$). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F).

The maximum value of N is _____.

gatecse-2018 co-and-architecture machine-instruction instruction-format numerical-answers two-marks

Answer key

1.18.7 Instruction Format: GATE CSE 2020 | Question: 44



A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.

gatecse-2020 co-and-architecture numerical-answers instruction-format machine-instruction two-marks

Answer key

1.18.8 Instruction Format: GATE CSE 2024 | Set 2 | Question: 47



A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____.

gatecse-2024-set2 numerical-answers co-and-architecture instruction-format two-marks

Answer key

1.18.9 Instruction Format: GATE CSE 2024 | Set 2 | Question: 51



A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.

R - type Instruction Format:

OPCODE	UNUSED	DSTRegister	SRCRegister1	SRCRegister2
--------	--------	-------------	--------------	--------------

I - type Instruction Format:

OPCODE	DSTRegister	SRCRegister	#Immediatevalue/address
--------	-------------	-------------	-------------------------

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.

Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field. The value of $X+2Y+Z$ is _____.

gatecse-2024-set2 numerical-answers co-and-architecture instruction-format two-marks

Answer key

1.18.10 Instruction Format: GATE CSE 2026 | Set 1 | Question: 5



Consider a processor P whose instruction set architecture is the load-store architecture. The instruction format is such that the first operand of any instruction is the destination operand.

Which one of the following sequences of instructions corresponds to the high-level language statement $Z = X + Y$?

Note: X, Y, and Z are memory operands. R0, R1, and R2 are registers.

- | | |
|-----------------------------|---|
| A. ADD Z, X, Y | B. LOAD R0, X ADD Z, R0, Y |
| C. ADD R0, X, Y STORE Z, R0 | D. LOAD R0, X LOAD R1, Y ADD R2, R0, R1 STORE Z, R2 |

gatecse-2026-set1 co-and-architecture instruction-format one-mark

Answer key

1.18.11 Instruction Format: GATE CSE 2026 | Set 2 | Question: 34



Consider a processor that has 16 general purpose registers and it uses 2-byte instruction format for all its instructions. Variable-sized opcodes are permitted. There are three different types of instructions; M-type, R-type, and C-type. Each M-type instruction has 2 register operands and a 6-bit immediate operand. Each Rtype instruction has 3 register operands. Each C-type instruction has a register operand and a 6-bit offset value. If there are 2 unique M-type opcodes and 7 unique R-type opcodes, which one of the following options gives the maximum

number of unique opcodes possible for C-type instructions?

- A. 8 B. 4 C. 64 D. 16

gatecse-2026-set2 co-and-architecture machine-instruction instruction-format two-marks

Answer key

1.19 Instruction Set Architecture (1)

1.19.1 Instruction Set Architecture: GATE CSE 2025 | Set 2 | Question: 18



Which of the following is/are part of an Instruction Set Architecture of a processor?

- A. The size of the cache memory B. The clock frequency of the processor
C. The number of cache memory levels D. The total number of registers

gatecse2025-set2 co-and-architecture instruction-set-architecture multiple-selects easy one-mark

Answer key

1.20 Interrupts (10)

Practice Test: Test 1 (11Q)

1.20.1 Interrupts: GATE CSE 1987 | Question: 1-viii



On receiving an interrupt from a I/O device the CPU:

- A. Halts for a predetermined time.
B. Hands over control of address bus and data bus to the interrupting device.
C. Branches off to the interrupt service routine immediately.
D. Branches off to the interrupt service routine after completion of the current instruction.

gate1987 co-and-architecture interrupts

Answer key

1.20.2 Interrupts: GATE CSE 1995 | Question: 1.3



In a vectored interrupt:

- A. The branch address is assigned to a fixed location in memory
B. The interrupting source supplies the branch information to the processor through an interrupt vector
C. The branch address is obtained from a register in the processor
D. None of the above

gate1995 co-and-architecture interrupts normal

Answer key

1.20.3 Interrupts: GATE CSE 1998 | Question: 1.20



Which of the following is true?

- A. Unless enabled, a CPU will not be able to process interrupts.
B. Loop instructions cannot be interrupted till they complete.
C. A processor checks for interrupts before executing a new instruction.
D. Only level triggered interrupts are possible on microprocessors.

gate1998 co-and-architecture interrupts normal

Answer key

1.20.4 Interrupts: GATE CSE 2002 | Question: 1.9



A device employing INTR line for device interrupt puts the CALL instruction on the data bus while:

- A. $\overline{INTA}$ is active
- B. HOLD is active
- C. READY is inactive
- D. None of the above

gatecse-2002 co-and-architecture interrupts normal

Answer key

1.20.5 Interrupts: GATE CSE 2005 | Question: 69



A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be $4\mu\text{sec}$. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode?

- A. 15
- B. 25
- C. 35
- D. 45

gatecse-2005 co-and-architecture interrupts

Answer key

1.20.6 Interrupts: GATE CSE 2009 | Question: 8, UGCNET-June2012-III: 58



A CPU generally handles an interrupt by executing an interrupt service routine:

- A. As soon as an interrupt is raised.
- B. By checking the interrupt register at the end of fetch cycle.
- C. By checking the interrupt register after finishing the execution of the current instruction.
- D. By checking the interrupt register at fixed time intervals.

gatecse-2009 co-and-architecture interrupts normal ugcnetcse-june2012-paper3

Answer key

1.20.7 Interrupts: GATE CSE 2020 | Question: 3



Consider the following statements.

- I. Daisy chaining is used to assign priorities in attending interrupts.
- II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt.
- III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.
- IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.

Which of the above statements is/are TRUE?

- A. I and II only
- B. I and IV only
- C. I and III only
- D. III only

gatecse-2020 co-and-architecture interrupts one-mark

Answer key

1.20.8 Interrupts: GATE CSE 2023 | Question: 24



A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes $100\mu\text{s}$ (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional $200\mu\text{s}$ to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke.

In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke.

The ratio $\frac{T_1}{T_2}$ is _____. (Rounded off to one decimal place)

gatecse-2023 co-and-architecture interrupts numerical-answers one-mark

Answer key

1.20.9 Interrupts: GATE CSE 2025 | Set 1 | Question: 1



Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives:

- The processor saves the content of the program counter.
- The program counter is loaded with the start address of the ISR.
- The processor finishes the present instruction.

Which ONE of the following is the CORRECT sequence of steps?

- A. (iii), (i), (ii) B. (i), (iii), (ii)
C. (i), (ii), (iii) D. (iii), (ii), (i)

gatecse2025-set1 co-and-architecture interrupts easy one-mark

Answer key

1.20.10 Interrupts: GATE CSE 2026 | Set 2 | Question: 8



Consider the following two statements about interrupt handling mechanisms in a CPU.

S1: In non-vectored interrupt mechanism, it usually takes more time to start the Interrupt Service Routine (ISR) when compared to that in a vectored interrupt mechanism.

S2: In daisy-chain interrupt mechanism, the CPU polls all the input devices individually to determine the source of the interrupt.

Which one of the following options is correct with respect to S1 and S2?

- A. Both S1 and S2 are true B. Both S1 and S2 are false
C. S1 is true and S2 is false D. S1 is false and S2 is true

gatecse-2026-set2 co-and-architecture interrupts one-mark

Answer key

1.21 Machine Instruction (21)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(8Q\)](#)

1.21.1 Machine Instruction: GATE CSE 1988 | Question: 9i



The following program fragment was written in an assembly language for a single address computer with one accumulator register:

```
LOAD B
MULT C
STORE T1
ADD A
STORE T2
MULT T2
ADD T1
STORE Z
```

Give the arithmetic expression implemented by the fragment.

gate1988 normal descriptive co-and-architecture machine-instruction

Answer key

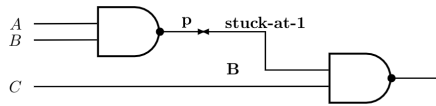
1.21.2 Machine Instruction: GATE CSE 1994 | Question: 12



- A. Assume that a CPU has only two registers R_1 and R_2 and that only the following instruction is available $XOR R_i, R_j; \{R_j \leftarrow R_i \oplus R_j, \text{ for } i, j = 1, 2\}$

Using this XOR instruction, find an instruction sequence in order to exchange the contents of the registers R_1 and R_2

- B. The line p of the circuit shown in figure has stuck at 1 fault. Determine an input test to detect the fault.



gate1994 co-and-architecture machine-instruction normal descriptive

Answer key

1.21.3 Machine Instruction: GATE CSE 1999 | Question: 17



Consider the following program fragment in the assembly language of a certain hypothetical processor. The processor has three general purpose registers $R1$, $R2$ and $R3$. The meanings of the instructions are shown by comments (starting with ;) after the instructions.

```
X: CMP R1, 0; Compare R1 and 0, set flags appropriately in status register
JZ Z; Jump if zero to target Z
MOV R2, R1; Copy contents of R1 to R2
SHR R1; Shift right R1 by 1 bit
SHL R1; Shift left R1 by 1 bit
CMP R2, R1; Compare R2 and R1 and set flag in status register
JZ Y; Jump if zero to target Y
INC R3; Increment R3 by 1;
Y: SHR R1; Shift right R1 by 1 bit
JMP X; Jump to target X
Z:...
```

- A. Initially $R1$, $R2$ and $R3$ contain the values 5, 0 and 0 respectively, what are the final values of $R1$ and $R3$ when control reaches Z?
- B. In general, if $R1$, $R2$ and $R3$ initially contain the values n, 0, and 0 respectively. What is the final value of $R3$ when control reaches Z?

gate1999 co-and-architecture machine-instruction normal descriptive

Answer key

1.21.4 Machine Instruction: GATE CSE 2003 | Question: 48



Consider the following assembly language program for a hypothetical processor A , B , and C are 8-bit registers. The meanings of various instructions are shown as comments.


```

MOV B, #0      ;  $B \leftarrow 0$ 
MOV C, #8      ;  $C \leftarrow 8$ 
Z:  CMP C, #0   ; compare C with 0
    JZ X        ; jump to X if zero flag is set
    SUB C, #1    ;  $C \leftarrow C - 1$ 
    RRC A, #1    ; right rotate A through carry by one bit. Thus:
                  ; If the initial values of A and the carry flag are  $a_7 \dots a_0$  and
                  ;  $c_0$  respectively, their values after the execution of this
                  ; instruction will be  $c_0 a_7 \dots a_1$  and  $a_0$  respectively.
    JC Y        ; jump to Y if carry flag is set
    JMP Z       ; jump to Z
Y:  ADD B, #1    ;  $B \leftarrow B + 1$ 
    JMP Z       ; jump to Z
X:

```

If the initial value of register A is A0 the value of register B after the program execution will be

- A. the number of 0 bits in A_0
- B. the number of 1 bits in A_0
- C. A_0
- D. 8

gatecse-2003 co-and-architecture machine-instruction normal

Answer key

1.21.5 Machine Instruction: GATE CSE 2003 | Question: 49



Consider the following assembly language program for a hypothetical processor A, B, and C are 8 bit registers. The meanings of various instructions are shown as comments.

```

MOV B, #0      ;  $B \leftarrow 0$ 
MOV C, #8      ;  $C \leftarrow 8$ 
Z:  CMP C, #0   ; compare C with 0
    JZ X        ; jump to X if zero flag is set
    SUB C, #1    ;  $C \leftarrow C - 1$ 
    RRC A, #1    ; right rotate A through carry by one bit. Thus:
                  ; If the initial values of A and the carry flag are  $a_7 \dots a_0$  and
                  ;  $c_0$  respectively, their values after the execution of this
                  ; instruction will be  $c_0 a_7 \dots a_1$  and  $a_0$  respectively.
    JC Y        ; jump to Y if carry flag is set
    JMP Z       ; jump to Z
Y:  ADD B, #1    ;  $B \leftarrow B + 1$ 
    JMP Z       ; jump to Z
X:

```

Which of the following instructions when inserted at location X will ensure that the value of the register A after program execution is as same as its initial value?

- A. RRC A, #1
- B. NOP; no operation
- C. LRC A, #1; left rotate A through carry flag by one bit
- D. ADD A, #1

gatecse-2003 co-and-architecture machine-instruction normal

Answer key

1.21.6 Machine Instruction: GATE CSE 2004 | Question: 63



Consider the following program segment for a hypothetical CPU having three user registers R_1 , R_2 and R_3 .

Instruction	Operation	Instruction size (in words)
MOV R_1 , 5000	$R_1 \leftarrow \text{Memory}[5000]$	2
MOV R_2 , (R_1)	$R_2 \leftarrow \text{Memory}[(R_1)]$	1
ADD R_2 , R_3	$R_2 \leftarrow R_2 + R_3$	1
MOV 6000, R_2	$\text{Memory}[6000] \leftarrow R_2$	2
HALT	Machine Halts	1

Consider that the memory is byte addressable with size 32 bits, and the program has been loaded starting from memory location 1000 (decimal). If an interrupt occurs while the CPU has been halted after executing the HALT instruction, the return address (in decimal) saved in the stack will be

- A. 1007 B. 1020 C. 1024 D. 1028

gatecse-2004 co-and-architecture machine-instruction normal

Answer key

1.21.7 Machine Instruction: GATE CSE 2004 | Question: 64



Consider the following program segment for a hypothetical CPU having three user registers R_1 , R_2 and R_3 .

Instruction	Operation	Instruction size (in Words)
MOV R_1 , 5000	$R_1 \leftarrow \text{Memory}[5000]$	2
MOV R_2 (R_1)	$R_2 \leftarrow \text{Memory}[(R_1)]$	1
ADD R_2 , R_3	$R_2 \leftarrow R_2 + R_3$	1
MOV 6000, R_2	$\text{Memory}[6000] \leftarrow R_2$	2
Halt	Machine Halts	1

Let the clock cycles required for various operations be as follows:

Register to/from memory transfer	3 clock cycles
ADD with both operands in register	1 clock cycles
Instruction fetch and decode	2 clock cycles

The total number of clock cycles required to execute the program is

- A. 29 B. 24 C. 23 D. 20

gatecse-2004 co-and-architecture machine-instruction normal

Answer key

1.21.8 Machine Instruction: GATE CSE 2006 | Question: 09, ISRO2009-35



A CPU has 24-bit instructions. A program starts at address 300 (in decimal). Which one of the following is a legal program counter (all values in decimal)?

- A. 400 B. 500 C. 600 D. 700

gatecse-2006 co-and-architecture machine-instruction easy isro2009

Answer key

1.21.9 Machine Instruction: GATE CSE 2007 | Question: 54



In a simplified computer the instructions are:

OP R_j, R_i	Perform R_j OP R_i and store the result in register R_j
OP m, R_i	Perform val OP R_i and store the result in register R_i val denotes the content of the memory location m
MOV m, R_i	Moves the content of memory location m to register R_i
MOV R_i, m	Moves the content of register R_i to memory location m

The computer has only two registers, and OP is either ADD or SUB. Consider the following basic block:

- $t_1 = a + b$
- $t_2 = c + d$
- $t_3 = e - t_2$
- $t_4 = t_1 - t_3$

Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

- A. 2 B. 3 C. 5 D. 6

gatecse-2007 co-and-architecture machine-instruction normal

Answer key

1.21.10 Machine Instruction: GATE CSE 2007 | Question: 71



Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	$R1 \leftarrow M[3000]$	2
LOOP:	MOV R2,(R3)	$R2 \leftarrow M[R3]$	1
	ADD R2,R1	$R2 \leftarrow R1 + R2$	1
	MOV (R3),R2	$M[R3] \leftarrow R2$	1
	INC R3	$R3 \leftarrow R3 + 1$	1
	DEC R1	$R1 \leftarrow R1 - 1$	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. The number of memory references for accessing the data in executing the program completely is

- A. 10 B. 11 C. 20 D. 21

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.11 Machine Instruction: GATE CSE 2007 | Question: 72



Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	$R1 \leftarrow M[3000]$	2
LOOP:	MOV R2,(R3)	$R2 \leftarrow M[R3]$	1
	ADD R2,R1	$R2 \leftarrow R1 + R2$	1
	MOV (R3),R2	$M[R3] \leftarrow R2$	1
	INC R3	$R3 \leftarrow R3 + 1$	1
	DEC R1	$R1 \leftarrow R1 - 1$	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is:

- A. 100 B. 101 C. 102 D. 110

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.12 Machine Instruction: GATE CSE 2007 | Question: 73



Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction Size (no. of words)
	MOV R1,(3000)	$R1 \leftarrow M[3000]$	2
LOOP:	MOV R2,(R3)	$R2 \leftarrow M[R3]$	1
	ADD R2,R1	$R2 \leftarrow R1 + R2$	1
	MOV (R3),R2	$M[R3] \leftarrow R2$	1
	INC R3	$R3 \leftarrow R3 + 1$	1
	DEC R1	$R1 \leftarrow R1 - 1$	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?

- A. 1005 B. 1020 C. 1024 D. 1040

gatecse-2007 co-and-architecture machine-instruction interrupts normal

Answer key

1.21.13 Machine Instruction: GATE CSE 2008 | Question: 34



Which of the following must be true for the RFE (Return From Exception) instruction on a general purpose processor?

- I. It must be a trap instruction
- II. It must be a privileged instruction
- III. An exception cannot be allowed to occur during execution of an RFE instruction

A. I only

B. II only

C. I and II only

D. I, II and III only

gatecse-2008 co-and-architecture machine-instruction normal

Answer key

1.21.14 Machine Instruction: GATE CSE 2015 | Set 2 | Question: 42



Consider a processor with byte-addressable memory. Assume that all registers, including program counter (PC) and Program Status Word (PSW), are size of two bytes. A stack in the main memory is implemented from memory location $(0100)_{16}$ and it grows upward. The stack pointer (SP) points to the top element of the stack. The current value of SP is $(016E)_{16}$. The CALL instruction is of two words, the first word is the op-code and the second word is the starting address of the subroutine (one word = 2 bytes). The CALL instruction is implemented as follows:

- Store the current value of PC in the stack
- Store the value of PSW register in the stack
- Load the starting address of the subroutine in PC

The content of PC just before the fetch of a CALL instruction is $(5FA0)_{16}$. After execution of the CALL instruction, the value of the stack pointer is:

A. $(016A)_{16}$ B. $(016C)_{16}$ C. $(0170)_{16}$ D. $(0172)_{16}$

gatecse-2015-set2 co-and-architecture machine-instruction easy

Answer key

1.21.15 Machine Instruction: GATE CSE 2016 | Set 2 | Question: 10



A processor has 40 distinct instruction and 24 general purpose registers. A 32-bit instruction word has an opcode, two registers operands and an immediate operand. The number of bits available for the immediate operand field is _____.

gatecse-2016-set2 machine-instruction co-and-architecture easy numerical-answers

Answer key

1.21.16 Machine Instruction: GATE CSE 2021 | Set 1 | Question: 55



Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY[X] denotes the content at the memory location X.

Instruction	Semantics	Instruction Size (bytes)
MOV R1, (5000)	$R1 \leftarrow \text{MEMORY}[5000]$	4
MOV R2, (R3)	$R2 \leftarrow \text{MEMORY}[R3]$	4
ADDR2, R1	$R2 \leftarrow R1 + R2$	2
MOV (R3), R2	$\text{MEMORY}[R3] \leftarrow R2$	4
INC R3	$R3 \leftarrow R3 + 1$	2
DEC R1	$R1 \leftarrow R1 - 1$	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3020 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is _____

gatecse-2021-set1 co-and-architecture machine-instruction numerical-answers two-marks

Answer key

1.21.17 Machine Instruction: GATE CSE 2023 | Question: 31



Consider the given C-code and its corresponding assembly code, with a few operands U1-U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable.

<pre>//C-code int a[10], b[10], i; // int is 32 bit for(i=0; i<10; i++) a[i] = b[i] * 8;</pre>	<pre>;assembly code (; indicates comments) ;r1-r5 are 32-bit integer registers ;initialize r1=0, r2=10 ;initialize r3, r4 with base address of a, b L01: jeq r1, r2, end ;if(r1==r2) goto end L02: lw, r5, 0(r4) ;r5 <- Memory[r4+0] L03: shl, r5, r5, U1 ;r5 <- r5 << U1 L04: sw, r5, 0(r3) ;Memory[r3+0] <- r5 L05: add, r3, r3, U2 ;r3 <- r3+U2 L06: add, r4, r4, U3 L07: add, r1, r1, 1 L08: jmp U4 ;goto U4 L09: end</pre>
---	--

Which one of the following options is a **CORRECT** replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?

- A. (8,4,1,L02) B. (3,4,4,L01) C. (8,1,1,L02) D. (3,1,1,L01)

gatecse-2023 co-and-architecture machine-instruction two-marks

Answer key

1.21.18 Machine Instruction: GATE CSE 2025 | Set 1 | Question: 27



A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction?

ADD R1, #25 // R1 = R1 + 25

- A. 16 B. 20 C. 22 D. 24

gatecse2025-set1 co-and-architecture machine-instruction easy two-marks

Answer key

1.21.19 Machine Instruction: GATE IT 2004 | Question: 46



If we use internal data forwarding to speed up the performance of a CPU (R1, R2 and R3 are registers and M[100] is a memory reference), then the sequence of operations

R1 → M[100]

M[100] → R2

M[100] → R3

can be replaced by

- A. R1 → R3
R2 → M[100]
- B. M[100] → R2
R1 → R2
R1 → R3
- C. R1 → M[100]
R2 → R3
- D. R1 → R2
R1 → R3
R1 → M[100]

gateit-2004 co-and-architecture machine-instruction easy

Answer key

1.21.20 Machine Instruction: GATE IT 2007 | Question: 41



Following table indicates the latencies of operations between the instruction producing the result and

instruction using the result.

Instruction producing the result	Instruction using the result	Latency
ALU Operation	ALU Operation	2
ALU Operation	Store	2
Load	ALU Operation	1
Load	Store	0

Consider the following code segment:

```
Load R1, Loc 1; Load R1 from memory location Loc1
Load R2, Loc 2; Load R2 from memory location Loc 2
Add R1, R2, R1; Add R1 and R2 and save result in R1
Dec R2; Decrement R2
Dec R1; Decrement R1
Mpy R1, R2, R3; Multiply R1 and R2 and save result in R3
Store R3, Loc 3; Store R3 in memory location Loc 3
```

What is the number of cycles needed to execute the above code segment assuming each instruction takes one cycle to execute?

- A. 7 B. 10 C. 13 D. 14

gateit-2007 co-and-architecture machine-instruction normal

Answer key

1.21.21 Machine Instruction: GATE IT 2008 | Question: 38



Assume that $EA = (X)+$ is the effective address equal to the contents of location X, with X incremented by one word length after the effective address is calculated; $EA = -(X)$ is the effective address equal to the contents of location X, with X decremented by one word length before the effective address is calculated; $EA = (X)-$ is the effective address equal to the contents of location X, with X decremented by one word length after the effective address is calculated. The format of the instruction is (opcode, source, destination), which means (destination $\leftarrow$ source op destination). Using X as a stack pointer, which of the following instructions can pop the top two elements from the stack, perform the addition operation and push the result back to the stack.

- A. ADD $(X)-, (X)$ B. ADD $(X), (X)-$
C. ADD $-(X), (X)+$ D. ADD $-(X), (X)$

gateit-2008 co-and-architecture machine-instruction normal

Answer key

1.22 Memory Interfacing (6)

Practice Test: Test 1 (10Q)

1.22.1 Memory Interfacing: GATE CSE 1990 | Question: 4-iv



State whether the following statements are TRUE or FALSE with reason:

Transferring data in blocks from the main memory to the cache memory enables an interleaved main memory unit to operate at its maximum speed.

gate1990 true-false co-and-architecture cache-memory memory-interfacing

Answer key

1.22.2 Memory Interfacing: GATE CSE 1991 | Question: 01,ii



In interleaved memory organization, consecutive words are stored in consecutive memory modules in _____ interleaving, whereas consecutive words are stored within the module in _____ interleaving.

gate1991 co-and-architecture normal memory-interfacing descriptive

Answer key

1.22.3 Memory Interfacing: GATE CSE 2006 | Question: 41



A CPU has a cache with block size 64 bytes. The main memory has k banks, each bank being c bytes wide. Consecutive c – byte chunks are mapped on consecutive banks with wrap-around. All the k banks can be accessed in parallel, but two accesses to the same bank must be serialized. A cache block access may involve multiple iterations of parallel bank accesses depending on the amount of data obtained by accessing all the k banks in parallel. Each iteration requires decoding the bank numbers to be accessed in parallel and this takes $\frac{k}{2}ns$. The latency of one bank access is 80 ns. If $c = 2$ and $k = 24$, the latency of retrieving a cache block starting at address zero from main memory is:

- A. 92 ns B. 104 ns C. 172 ns D. 184 ns

gatecse-2006 co-and-architecture cache-memory memory-interfacing normal

Answer key

1.22.4 Memory Interfacing: GATE CSE 2016 | Set 1 | Question: 09



A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of address bus of the processor is at least _____ bits.

gatecse-2016-set1 co-and-architecture easy numerical-answers memory-interfacing

Answer key

1.22.5 Memory Interfacing: GATE CSE 2018 | Question: 23



A 32-bit wide main memory unit with a capacity of 1 GB is built using $256M \times 4$ -bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14} . The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.

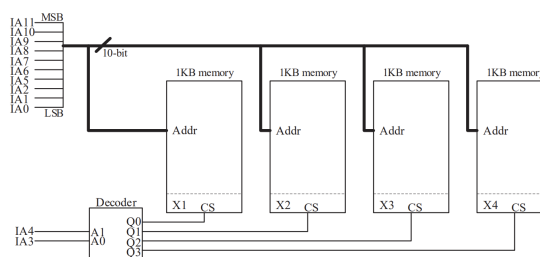
gatecse-2018 co-and-architecture memory-interfacing normal numerical-answers one-mark

Answer key

1.22.6 Memory Interfacing: GATE CSE 2023 | Question: 32



A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11-IA0 are connected to the address port of these blocks. The chip select (CS) is active high.



The input memory addresses (IA11-IA0), in decimal, for the starting locations (Addr = 0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

- A. (0,1,2,3) B. (0,1024,2048,3072) C. (0,8,16,24) D. (0,0,0,0)

gatecse-2023 co-and-architecture memory-interfacing two-marks

Answer key

1.23

Microprogramming (12)

Practice Test: Test 1 (15Q)

1.23.1 Microprogramming: GATE CSE 1987 | Question: 4a



Find out the width of the control memory of a horizontal microprogrammed control unit, given the following specifications:

- 16 control lines for the processor consisting of ALU and 7 registers.
- Conditional branching facility by checking 4 status bits.
- Provision to hold 128 words in the control memory.

gate1987 co-and-architecture microprogramming descriptive

Answer key

1.23.2 Microprogramming: GATE CSE 1996 | Question: 2.25



A micro program control unit is required to generate a total of 25 control signals. Assume that during any micro instruction, at most two control signals are active. Minimum number of bits required in the control word to generate the required control signals will be:

- A. 2 B. 2.5 C. 10 D. 12

gate1996 co-and-architecture microprogramming normal

Answer key

1.23.3 Microprogramming: GATE CSE 1997 | Question: 5.3



A micro instruction is to be designed to specify:

- a. none or one of the three micro operations of one kind and
b. none or upto six micro operations of another kind

The minimum number of bits in the micro-instruction is:

- A. 9 B. 5 C. 8 D. None of the above

gate1997 co-and-architecture microprogramming normal

Answer key

1.23.4 Microprogramming: GATE CSE 1999 | Question: 2.19



Arrange the following configuration for CPU in decreasing order of operating speeds:
Hard wired control, Vertical microprogramming, Horizontal microprogramming.

- A. Hard wired control, Vertical microprogramming, Horizontal microprogramming.
B. Hard wired control, Horizontal microprogramming, Vertical microprogramming.
C. Horizontal microprogramming, Vertical microprogramming, Hard wired control.
D. Vertical microprogramming, Horizontal microprogramming, Hard wired control.

gate1999 co-and-architecture microprogramming normal

Answer key

1.23.5 Microprogramming: GATE CSE 2002 | Question: 2.7



Horizontal microprogramming:

- A. does not require use of signal decoders
B. results in larger sized microinstructions than vertical microprogramming
C. uses one bit for each control signal
D. all of the above

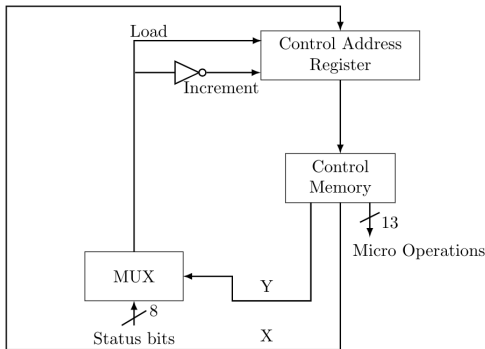
gatecse-2002 co-and-architecture microprogramming

Answer key

1.23.6 Microprogramming: GATE CSE 2004 | Question: 67



The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the input of the MUX.



How many bits are there in the X and Y fields, and what is the size of the control memory in number of words?

- A. 10,3,1024 B. 8,5,256 C. 5,8,2048 D. 10,3,512

gatecse-2004 co-and-architecture microprogramming normal

Answer key

1.23.7 Microprogramming: GATE CSE 2013 | Question: 28



Consider the following sequence of micro-operations.

```
MBR ← PC
MAR ← X PC ← Y
Memory ← MBR
```

Which one of the following is a possible operation performed by this sequence?

- A. Instruction fetch B. Operand fetch
C. Conditional branch D. Initiation of interrupt service

gatecse-2013 co-and-architecture microprogramming normal

Answer key

1.23.8 Microprogramming: GATE IT 2004 | Question: 49



A CPU has only three instructions $I1$, $I2$ and $I3$, which use the following signals in time steps $T1 - T5$:

$I1$: $T1$: Ain, Bout, Cin
 $T2$: PCout, Bin
 $T3$: Zout, Ain
 $T4$: Bin, Cout
 $T5$: End

$I2$: $T1$: Cin, Bout, Din
 $T2$: Aout, Bin
 $T3$: Zout, Ain
 $T4$: Bin, Cout
 $T5$: End

$I3$: $T1$: Din, Aout
 $T2$: Ain, Bout
 $T3$: Zout, Ain
 $T4$: Dout, Ain
 $T5$: End

Which of the following logic functions will generate the hardwired control for the signal Ain ?

- A. $T1.I1 + T2.I3 + T4.I3 + T3$ B. $(T1 + T2 + T3).I3 + T1.I1$
C. $(T1 + T2).I1 + (T2 + T4).I3 + T3$ D. $(T1 + T2).I2 + (T1 + T3).I1 + T3$

gateit-2004 co-and-architecture microprogramming normal

1.23.9 Microprogramming: GATE IT 2005 | Question: 45



A hardwired CPU uses 10 control signals S_1 to S_{10} , in various time steps T_1 to T_5 , to implement 4 instructions I_1 to I_4 as shown below:

	T_1	T_2	T_3	T_4	T_5
I_1	S_1, S_3, S_5	S_2, S_4, S_6	S_1, S_7	S_{10}	S_3, S_8
I_2	S_1, S_3, S_5	S_8, S_9, S_{10}	S_5, S_6, S_7	S_6	S_{10}
I_3	S_1, S_3, S_5	S_7, S_8, S_{10}	S_2, S_6, S_9	S_{10}	S_1, S_3
I_4	S_1, S_3, S_5	S_2, S_6, S_7	S_5, S_{10}	S_6, S_9	S_{10}

Which of the following pairs of expressions represent the circuit for generating control signals S_5 and S_{10} respectively?

$((I_j + I_k)T_n)$ indicates that the control signal should be generated in time step T_n if the instruction being executed is I_j or I_k

- A. $S_5 = T_1 + I_2 \cdot T_3$ and
 $S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$
- B. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and
 $S_{10} = (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$
- C. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and
 $S_{10} = (I_2 + I_3 + I_4) \cdot T_2 + (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$
- D. $S_5 = T_1 + (I_2 + I_4) \cdot T_3$ and
 $S_{10} = (I_2 + I_3) \cdot T_2 + I_4 \cdot T_3 + (I_1 + I_3) \cdot T_4 + (I_2 + I_4) \cdot T_5$

gateit-2005 co-and-architecture microprogramming normal

1.23.10 Microprogramming: GATE IT 2005 | Question: 49



An instruction set of a processor has 125 signals which can be divided into 5 groups of mutually exclusive signals as follows:

Group 1 : 20 signals, Group 2 : 70 signals, Group 3 : 2 signals, Group 4 : 10 signals, Group 5 : 23 signals.

How many bits of the control words can be saved by using vertical microprogramming over horizontal microprogramming?

- A. 0 B. 103 C. 22 D. 55

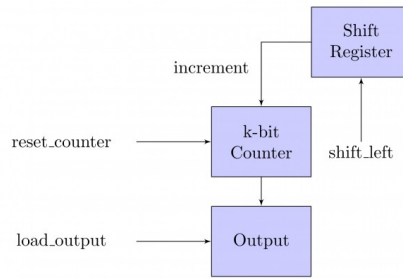
gateit-2005 co-and-architecture microprogramming normal

1.23.11 Microprogramming: GATE IT 2006 | Question: 41



The data path shown in the figure computes the number of 1s in the 32-bit input word corresponding to an unsigned even integer stored in the shift register.

The unsigned counter, initially zero, is incremented if the most significant bit of the shift register is 1.



The microprogram for the control is shown in the table below with missing control words for microinstructions $I_1, I_2, \dots, I_n$.

Microinstruction	Reset_Counter	Shift_left	Load_output
BEGIN	1	0	0
I_1	?	?	?
:	:	:	:
I_n	?	?	?
END	0	0	1

The counter width (k), the number of missing microinstructions (n), and the control word for microinstructions $I_1, I_2, \dots, I_n$ are, respectively,

- A. 32,5,010 B. 5,32,010 C. 5,31,011 D. 5,31,010

gateit-2006 co-and-architecture microprogramming normal

Answer key

1.23.12 Microprogramming: GATE IT 2008 | Question: 39



Consider a CPU where all the instructions require 7 clock cycles to complete execution. There are 140 instructions in the instruction set. It is found that 125 control signals are needed to be generated by the control unit. While designing the horizontal microprogrammed control unit, single address field format is used for branch control logic. What is the minimum size of the control word and control address register?

- A. 125,7 B. 125,10 C. 135,9 D. 135,10

gateit-2008 co-and-architecture microprogramming normal

Answer key

1.24 Pipelining (39)

Practice Tests: [Test 1 \(15Q\)](#) [Test 2 \(15Q\)](#) [Test 3 \(15Q\)](#) [Test 4 \(15Q\)](#) [Test 5 \(10Q\)](#)

1.24.1 Pipelining: GATE CSE 1999 | Question: 13



An instruction pipeline consists of 4 stages – Fetch (F), Decode field (D), Execute (E) and Result Write (W). The 5 instructions in a certain instruction sequence need these stages for the different number of clock cycles as shown by the table below

Instruction	F	D	E	W
1	1	2	1	1
2	1	2	2	1
3	2	1	3	2
4	1	3	2	1
5	1	2	1	2

Find the number of clock cycles needed to perform the 5 instructions.

gate1999 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.2 Pipelining: GATE CSE 2000 | Question: 1.8



Comparing the time T_1 taken for a single instruction on a pipelined CPU with time T_2 taken on a non-pipelined but identical CPU, we can say that

- A. $T_1 \leq T_2$
- B. $T_1 \geq T_2$
- C. $T_1 < T_2$
- D. T_1 and T_2 plus the time taken for one instruction fetch cycle

gatecse-2000 pipelining co-and-architecture easy

Answer key

1.24.3 Pipelining: GATE CSE 2000 | Question: 12



An instruction pipeline has five stages where each stage take 2 nanoseconds and all instruction use all five stages. Branch instructions are not overlapped. i.e., the instruction after the branch is not fetched till the branch instruction is completed. Under ideal conditions,

- A. Calculate the average instruction execution time assuming that 20% of all instructions executed are branch instruction. Ignore the fact that some branch instructions may be conditional.
- B. If a branch instruction is a conditional branch instruction, the branch need not be taken. If the branch is not taken, the following instructions can be overlapped. When 80% of all branch instructions are conditional branch instructions, and 50% of the conditional branch instructions are such that the branch is taken, calculate the average instruction execution time.

gatecse-2000 co-and-architecture pipelining normal descriptive

Answer key

1.24.4 Pipelining: GATE CSE 2001 | Question: 12



Consider a 5—stage pipeline - IF (Instruction Fetch), ID (Instruction Decode and register read), EX (Execute), MEM (memory), and WB (Write Back). All (memory or register) reads take place in the second phase of a clock cycle and all writes occur in the first phase. Consider the execution of the following instruction sequence:

I1:	sub $r2, r3, r4$	$/* \quad r2 \leftarrow r3 - r4 \quad */$
I2:	sub $r4, r2, r3$	$/* \quad r4 \leftarrow r2 - r3 \quad */$
I3:	sw $r2, 100(r1)$	$/* \quad M[r1 + 100] \leftarrow r2 \quad */$
I4:	sub $r3, r4, r2$	$/* \quad r3 \leftarrow r4 - r2 \quad */$

- A. Show all data dependencies between the four instructions.
- B. Identify the data hazards.
- C. Can all hazards be avoided by forwarding in this case.

gatecse-2001 co-and-architecture pipelining normal descriptive

Answer key

1.24.5 Pipelining: GATE CSE 2002 | Question: 2.6, ISRO2008-19



The performance of a pipelined processor suffers if:

- A. the pipeline stages have different delays
- B. consecutive instructions are dependent on each other
- C. the pipeline stages share hardware resources
- D. All of the above

Answer key

1.24.6 Pipelining: GATE CSE 2003 | Question: 10, ISRO-DEC2017-41



For a pipelined CPU with a single ALU, consider the following situations

- I. The $j + 1^{st}$ instruction uses the result of the j^{th} instruction as an operand
- II. The execution of a conditional jump instruction
- III. The j^{th} and $j + 1^{st}$ instructions require the ALU at the same time.

Which of the above can cause a hazard

- A. I and II only B. II and III only C. III only D. All the three

gatecse-2003 co-and-architecture pipelining normal isrodec2017

Answer key

1.24.7 Pipelining: GATE CSE 2004 | Question: 69



A 4-stage pipeline has the stage delays as 150, 120, 160 and 140 *nanoseconds*, respectively. Registers that are used between the stages have a delay of 5 *nanoseconds* each. Assuming constant clocking rate, the total time taken to process 1000 data items on this pipeline will be:

- A. 120.4 microseconds B. 160.5 microseconds
C. 165.5 microseconds D. 590.0 microseconds

gatecse-2004 co-and-architecture pipelining normal

Answer key

1.24.8 Pipelining: GATE CSE 2006 | Question: 42



A CPU has a five-stage pipeline and runs at 1 GHz frequency. Instruction fetch happens in the first stage of the pipeline. A conditional branch instruction computes the target address and evaluates the condition in the third stage of the pipeline. The processor stops fetching new instructions following a conditional branch until the branch outcome is known. A program executes 10^9 instructions out of which 20% are conditional branches. If each instruction takes one cycle to complete on average, the total execution time of the program is:

- A. 1.0 second B. 1.2 seconds C. 1.4 seconds D. 1.6 seconds

gatecse-2006 co-and-architecture pipelining normal

Answer key

1.24.9 Pipelining: GATE CSE 2007 | Question: 37, ISRO2009-37



Consider a pipelined processor with the following four stages:

- IF: Instruction Fetch
- ID: Instruction Decode and Operand Fetch
- EX: Execute
- WB: Write Back

The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD	R2, R1, R0	$R2 \leftarrow R1 + R0$
MUL	R4, R3, R2	$R4 \leftarrow R3 * R2$
SUB	R6, R5, R4	$R6 \leftarrow R5 - R4$

- A. 7 B. 8 C. 10 D. 14

Answer key

1.24.10 Pipelining: GATE CSE 2008 | Question: 76



Delayed branching can help in the handling of control hazards

For all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false,

- A. The instruction following the conditional branch instruction in memory is executed
- B. The first instruction in the fall through path is executed
- C. The first instruction in the taken path is executed
- D. The branch takes longer to execute than any other instruction

Answer key

1.24.11 Pipelining: GATE CSE 2008 | Question: 77



Delayed branching can help in the handling of control hazards

The following code is to run on a pipelined processor with one branch delay slot:

I1: $ADD\ R2 \leftarrow R7 + R8$

I2: $Sub\ R4 \leftarrow R5 - R6$

I3: $ADD\ R1 \leftarrow R2 + R3$

I4: $STORE\ Memory\ [R4] \leftarrow R1$

BRANCH to Label if $R1 == 0$

Which of the instructions I1, I2, I3 or I4 can legitimately occupy the delay slot without any program modification?

- A. I1
- B. I2
- C. I3
- D. I4

Answer key

1.24.12 Pipelining: GATE CSE 2009 | Question: 28



Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions $I1, I2, I3, I4$ in stages $S1, S2, S3, S4$ is shown below:

	S1	S2	S3	S4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop?

For (i=1 to 2) {I1; I2; I3; I4;}

- A. 16
- B. 23
- C. 28
- D. 30

Answer key

1.24.13 Pipelining: GATE CSE 2010 | Question: 33



A 5—stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, OF and WO stages take 1 clock cycle

each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?

Instruction	Meaning of instruction
t_0 : MUL R_2, R_0, R_1	$R_2 \leftarrow R_0 * R_1$
t_1 : DIV R_5, R_3, R_4	$R_5 \leftarrow R_3 / R_4$
t_2 : ADD R_2, R_5, R_2	$R_2 \leftarrow R_5 + R_2$
t_3 : SUB R_5, R_2, R_6	$R_5 \leftarrow R_2 - R_6$

- A. 13 B. 15 C. 17 D. 19

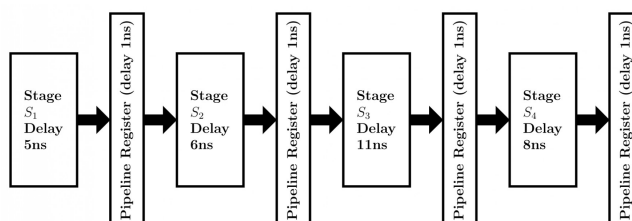
gatecse-2010 co-and-architecture pipelining normal

Answer key

1.24.14 Pipelining: GATE CSE 2011 | Question: 41



Consider an instruction pipeline with four stages (S_1, S_2, S_3 and S_4) each with combinational circuit only. The pipeline registers are required between each stage and at the end of the last stage. Delays for the stages and for the pipeline registers are as given in the figure.



What is the approximate speed up of the pipeline in steady state under ideal conditions when compared to the corresponding non-pipeline implementation?

- A. 4.0 B. 2.5 C. 1.1 D. 3.0

gatecse-2011 co-and-architecture pipelining normal

Answer key

1.24.15 Pipelining: GATE CSE 2012 | Question: 20, ISRO2016-23



Register renaming is done in pipelined processors:

- A. as an alternative to register allocation at compile time
- B. for efficient access to function parameters and local variables
- C. to handle certain kinds of hazards
- D. as part of address translation

gatecse-2012 co-and-architecture pipelining easy isro2016

Answer key

1.24.16 Pipelining: GATE CSE 2013 | Question: 45



Consider an instruction pipeline with five stages without any branch prediction:

Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions $I_1, I_2, I_3, \dots, I_{12}$ is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_9 . If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

- A. 132 B. 165
C. 176 D. 328

Answer key

1.24.17 Pipelining: GATE CSE 2014 | Set 3 | Question: 43



An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (EX), memory access (MEM), and register writeback (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively (ns stands for nanoseconds). To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 2.2/3 ns. Also, the EX stage is split into two stages (EX1, EX2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the EX stage and produce the next instruction pointer at the end of the EX stage in the old design and at the end of the EX2 stage in the new design. The IF stage stalls after fetching a branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is _____.

gatecse-2014-set3 co-and-architecture pipelining numerical-answers normal

Answer key

1.24.18 Pipelining: GATE CSE 2014 | Set 3 | Question: 9



Consider the following processors (ns stands for nanoseconds). Assume that the pipeline registers have zero latency.

- P1: Four-stage pipeline with stage latencies 1 ns, 2 ns, 2 ns, 1 ns.
- P2: Four-stage pipeline with stage latencies 1 ns, 1.5 ns, 1.5 ns, 1.5 ns.
- P3: Five-stage pipeline with stage latencies 0.5 ns, 1 ns, 1 ns, 0.6 ns, 1 ns.
- P4: Five-stage pipeline with stage latencies 0.5 ns, 0.5 ns, 1 ns, 1 ns, 1.1 ns.

Which processor has the highest peak clock frequency?

- A. P1 B. P2 C. P3 D. P4

gatecse-2014-set3 co-and-architecture pipelining normal

Answer key

1.24.19 Pipelining: GATE CSE 2015 | Set 1 | Question: 38



Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speedup achieved in this pipelined processor is _____.

gatecse-2015-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.20 Pipelining: GATE CSE 2015 | Set 2 | Question: 44



Consider the sequence of machine instruction given below:

```
MUL  R5, R0, R1
DIV   R6, R2, R3
ADD   R7, R5, R6
SUB   R8, R7, R4
```

In the above sequence, $R0$ to $R8$ are general purpose registers. In the instructions shown, the first register shows the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following 4 stages: (1) Instruction Fetch and Decode (IF), (2) Operand Fetch (OF), (3) Perform Operation (PO) and (4) Write back the result (WB). The IF , OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses

operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instruction is _____.

gatecse-2015-set2 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.21 Pipelining: GATE CSE 2015 | Set 3 | Question: 51



Consider the following reservation table for a pipeline having three stages S_1 , S_2 and S_3 .

Time →					
	1	2	3	4	5
S_1	X				X
S_2		X		X	
S_3			X		

The minimum average latency (MAL) is _____

gatecse-2015-set3 co-and-architecture pipelining difficult numerical-answers

Answer key

1.24.22 Pipelining: GATE CSE 2016 | Set 1 | Question: 32



The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionality equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is _____ percent.

gatecse-2016-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.23 Pipelining: GATE CSE 2016 | Set 2 | Question: 33



Consider a 3 GHz (gigahertz) processor with a three stage pipeline and stage latencies τ_1 , τ_2 and τ_3 such that $\tau_1 = \frac{3\tau_2}{4} = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.

gatecse-2016-set2 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.24 Pipelining: GATE CSE 2017 | Set 1 | Question: 50



Instruction execution in a processor is divided into 5 stages, *Instruction Fetch* (IF), *Instruction Decode* (ID), *Operand fetch* (OF), *Execute* (EX), and *Write Back* (WB). These stages take 5, 4, 20, 10 and 3 nanoseconds (ns) respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of 2 ns. Two pipelined implementation of the processor are contemplated:

- a naive pipeline implementation (NP) with 5 stages and
- an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of 12 ns and 8 ns respectively.

The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is _____.

gatecse-2017-set1 co-and-architecture pipelining normal numerical-answers

Answer key

1.24.25 Pipelining: GATE CSE 2018 | Question: 50



The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (*IF*), Instruction Decode (*ID*), Operand Fetch (*OF*), Perform Operation (*PO*) and Writeback (*WB*). The *IF*, *ID*, *OF* and *WB* stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the *PO* stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instruction is _____.

gatescse-2018 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.26 Pipelining: GATE CSE 2020 | Question: 43



Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5- stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____.

gatescse-2020 numerical-answers co-and-architecture pipelining two-marks

Answer key

1.24.27 Pipelining: GATE CSE 2021 | Set 1 | Question: 53



A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each. The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds.

gatescse-2021-set1 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.28 Pipelining: GATE CSE 2021 | Set 2 | Question: 53



Consider a pipelined processor with 5 stages, Instruction Fetch(*IF*), Instruction Decode(*ID*), Execute (*EX*), Memory Access (*MEM*), and Write Back (*WB*). Each stage of the pipeline, except the *EX* stage, takes one cycle. Assume that the *ID* stage merely decodes the instruction and the register read is performed in the *EX* stage. The *EX* stage takes one cycle for *ADD* instruction and two cycles for *MUL* instruction. Ignore pipeline register latencies.

Consider the following sequence of 8 instructions:

ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL

Assume that every *MUL* instruction is data-dependent on the *ADD* instruction just before it and every *ADD* instruction (except the first *ADD*) is data-dependent on the *MUL* instruction just before it. The *speedup* defined as follows.

$$Speedup = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}}$$

The *Speedup* achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____

gatescse-2021-set2 co-and-architecture pipelining numerical-answers two-marks

Answer key

1.24.29 Pipelining: GATE CSE 2023 | Question: 23



Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle.

The total execution time for executing 100 instructions on this processor is _____ ns.

gatecse-2023 co-and-architecture pipelining numerical-answers one-mark

Answer key

1.24.30 Pipelining: GATE CSE 2024 | Set 1 | Question: 20



Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about forwarding is/are CORRECT?

- A. In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction
- B. In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction
- C. Forwarding cannot prevent all pipeline stalls
- D. Forwarding does not require any extra hardware to retrieve the data from the pipeline stages

gatecse-2024-set1 multiple-selects co-and-architecture pipelining one-mark

Answer key

1.24.31 Pipelining: GATE CSE 2024 | Set 2 | Question: 21



An instruction format has the following structure:

Instruction Number: *Opcode destination reg, source reg-1, source reg-2*

Consider the following sequence of instructions to be executed in a pipelined processor:

I 1: DIV R3, R1, R2
I 2: SUB R5, R3, R4
I 3: ADD R3, R5, R6
I 4: MUL R7, R3, R8

Which of the following statements is/are TRUE?

- A. There is a RAW dependency on R 3 between I 1 and I 2
- B. There is a WAR dependency on R 3 between I 1 and I 3
- C. There is a RAW dependency on R 3 between I 2 and I 3
- D. There is a WAW dependency on R 3 between I 3 and I 4

gatecse-2024-set2 co-and-architecture multiple-selects pipelining one-mark

Answer key

1.24.32 Pipelining: GATE CSE 2024 | Set 2 | Question: 48



A non-pipelined instruction execution unit operating at 2GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (rounded off to one decimal place) obtained by the pipelined design over the non-pipelined design is _____.

gatecse-2024-set2 numerical-answers co-and-architecture pipelining two-marks

Answer key

1.24.33 Pipelining: GATE CSE 2025 | Set 2 | Question: 46



A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____. (rounded off to two decimal places)

gatecse2025-set2 co-and-architecture pipelining numerical-answers easy two-marks

Answer key

1.24.34 Pipelining: GATE IT 2004 | Question: 47



Consider a pipeline processor with 4 stages S_1 to S_4 . We want to execute the following loop:

```
for (i = 1; i <= 1000; i++)
{ I1, I2, I3, I4 }
```

where the time taken (in ns) by instructions I_1 to I_4 for stages S_1 to S_4 are given below:

	S_1	S_2	S_3	S_4
I1	1	2	1	2
I2	2	1	2	1
I3	1	1	2	1
I4	2	1	2	1

The output of I_1 for $i = 2$ will be available after

- A. 11 ns B. 12 ns C. 13 ns D. 28 ns

gateit-2004 co-and-architecture pipelining normal

Answer key

1.24.35 Pipelining: GATE IT 2005 | Question: 44



We have two designs D_1 and D_2 for a synchronous pipeline processor. D_1 has 5 pipeline stages with execution times of 3 nsec, 2 nsec, 4 nsec, 2 nsec and 3 nsec while the design D_2 has 8 pipeline stages each with 2 nsec execution time. How much time can be saved using design D_2 over design D_1 for executing 100 instructions?

- A. 214 nsec B. 202 nsec
C. 86 nsec D. -200 nsec

gateit-2005 co-and-architecture pipelining normal

Answer key

1.24.36 Pipelining: GATE IT 2006 | Question: 78



A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q)) / T$ is given below. The values of variables P, Q, R, S and T are available in the registers R_0, R_1, R_2, R_3 and R_4 respectively, before the execution of the instruction sequence.

```
ADD    R5, R0, R1    ; R5 ← R0 + R1
MUL    R6, R2, R5    ; R6 ← R2 * R5
SUB    R5, R3, R6    ; R5 ← R3 - R6
DIV    R6, R5, R4    ; R6 ← R5 / R4
STORE  R6, X        ; X ← R6
```

The number of Read-After-Write (RAW) dependencies, Write-After-Read (WAR) dependencies, and Write-After-Write (WAW) dependencies in the sequence of instructions are, respectively,

A. 2,2,4

B. 3,2,3

C. 4,2,2

D. 3,3,2

gateit-2006 co-and-architecture pipelining normal

Answer key

1.24.37 Pipelining: GATE IT 2006 | Question: 79



A pipelined processor uses a 4-stage instruction pipeline with the following stages: Instruction fetch (IF), Instruction decode (ID), Execute (EX) and Writeback (WB). The arithmetic operations as well as the load and store operations are carried out in the EX stage. The sequence of instructions corresponding to the statement $X = (S - R * (P + Q)) / T$ is given below. The values of variables P, Q, R, S and T are available in the registers $R0, R1, R2, R3$ and $R4$ respectively, before the execution of the instruction sequence.

```

ADD    R5, R0, R1 ; R5 ← R0 + R1
MUL    R6, R2, R5 ; R6 ← R2 * R5
SUB    R5, R3, R6 ; R5 ← R3 - R6
DIV    R6, R5, R4 ; R6 ← R5 / R4
STORE  R6, X      ; X ← R6

```

The IF, ID and WB stages take 1 clock cycle each. The EX stage takes 1 clock cycle each for the ADD, SUB and STORE operations, and 3 clock cycles each for MUL and DIV operations. Operand forwarding from the EX stage to the ID stage is used. The number of clock cycles required to complete the sequence of instructions is

A. 10

B. 12

C. 14

D. 16

gateit-2006 co-and-architecture pipelining normal

Answer key

1.24.38 Pipelining: GATE IT 2007 | Question: 6, ISRO2011-25



A processor takes 12 cycles to complete an instruction I . The corresponding pipelined processor uses 6 stages with the execution times of 3, 2, 5, 4, 6 and 2 cycles respectively. What is the asymptotic speedup assuming that a very large number of instructions are to be executed?

A. 1.83

B. 2

C. 3

D. 6

gateit-2007 co-and-architecture pipelining normal isro2011

Answer key

1.24.39 Pipelining: GATE IT 2008 | Question: 40



A non pipelined single cycle processor operating at 100 MHz is converted into a synchronous pipelined processor with five stages requiring 2.5 nsec, 1.5 nsec, 2 nsec, 1.5 nsec and 2.5 nsec, respectively. The delay of the latches is 0.5 nsec. The speedup of the pipeline processor for a large number of instructions is:

A. 4.5

B. 4.0

C. 3.33

D. 3.0

gateit-2008 co-and-architecture pipelining normal

Answer key

1.25

Runtime Environment (2)

1.25.1 Runtime Environment: GATE CSE 2001 | Question: 1.10, UGCNET-Dec2012-III: 36



Suppose a processor does not have any stack pointer registers, which of the following statements is true?

- A. It cannot have subroutine call instruction
- B. It cannot have nested subroutines call
- C. Interrupts are not possible
- D. All subroutine calls and interrupts are possible

gatecse-2001 co-and-architecture normal ugcnetcse-dec2012-paper3 runtime-environment

Answer key

1.25.2 Runtime Environment: GATE CSE 2008 | Question: 37, ISRO2009-38



The use of multiple register windows with overlap causes a reduction in the number of memory accesses for:

- I. Function locals and parameters
- II. Register saves and restores
- III. Instruction fetches

- A. I only B. II only C. III only D. I, II and III

gatecse-2008 co-and-architecture normal isro2009 runtime-environment

Answer key

1.26

Speedup (6)

Practice Test: Test 1 (11Q)

1.26.1 Speedup: GATE CSE 2014 | Set 1 | Question: 43



Consider a 6-stage instruction pipeline, where all stages are perfectly balanced. Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution if 25% of the instructions incur 2 pipeline stall cycles is _____.

gatecse-2014-set1 co-and-architecture pipelining numerical-answers normal speedup

Answer key

1.26.2 Speedup: GATE CSE 2014 | Set 1 | Question: 55



Consider two processors P_1 and P_2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P_2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P_1 . If the clock frequency of P_1 is 1GHz, then the clock frequency of P_2 (in GHz) is _____.

gatecse-2014-set1 co-and-architecture numerical-answers normal speedup

Answer key

1.26.3 Speedup: GATE CSE 2024 | Set 1 | Question: 45



The baseline execution time of a program on a 2GHz single core machine is 100 nanoseconds (ns). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 ns when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is _____.

gatecse-2024-set1 numerical-answers co-and-architecture speedup two-marks

Answer key

1.26.4 Speedup: GATE CSE 2026 | Set 2 | Question: 47



A non-pipelined instruction execution unit that operates at 1.6 GHz clock takes an average of 5 clock cycles to complete the execution of an instruction. To improve the performance, the system was pipelined with a goal of achieving an average throughput of one instruction per clock cycle. However, it could operate only at 1.2 GHz due to pipeline overheads. While executing a program in the pipelined design, 30% of instructions encountered a stall of 2 cycles due to pipeline hazards. The speed-up obtained by the pipelined design over the non-pipelined one for this program is _____ (rounded off to two decimal places)

Note: $1G = 10^9$

gatecse-2026-set2 co-and-architecture pipelining speedup numerical-answers two-marks

Answer key

1.26.5 Speedup: GATE IT 2004 | Question: 50



In an enhancement of a design of a CPU, the speed of a floating point unit has been increased by 20% and the speed of a fixed point unit has been increased by 10%. What is the overall speedup achieved if the ratio of the number of floating point operations to the number of fixed point operations is 2 : 3 and the floating point operation used to take twice the time taken by the fixed point operation in the original design?

- A. 1.155 B. 1.185 C. 1.255 D. 1.285

gateit-2004 normal co-and-architecture speedup

Answer key

1.26.6 Speedup: GATE IT 2007 | Question: 36



The floating point unit of a processor using a design D takes $2t$ cycles compared to t cycles taken by the fixed point unit. There are two more design suggestions D_1 and D_2 . D_1 uses 30% more cycles for fixed point unit but 30% less cycles for floating point unit as compared to design D . D_2 uses 40% less cycles for fixed point unit but 10% more cycles for floating point unit as compared to design D . For a given program which has 80% fixed point operations and 20% floating point operations, which of the following ordering reflects the relative performances of three designs?

($D_i > D_j$ denotes that D_i is faster than D_j)

- A. $D_1 > D > D_2$ B. $D_2 > D > D_1$
C. $D > D_2 > D_1$ D. $D > D_1 > D_2$

gateit-2007 co-and-architecture normal speedup

Answer key

1.27 Stall (1)

1.27.1 Stall: GATE CSE 2022 | Question: 51



A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2 . If the BPU has a prediction accuracy of 80%, the speed up (*rounded off to two decimal places*) obtained by X_2 over X_1 in executing P is _____.

gatecse-2022 numerical-answers co-and-architecture pipelining stall two-marks

Answer key

1.28 Virtual Memory (3)

1.28.1 Virtual Memory: GATE CSE 1991 | Question: 03,iii



The total size of address space in a virtual memory system is limited by:

- A. the length of MAR B. the available secondary storage
C. the available main memory D. all of the above
E. none of the above

gate1991 co-and-architecture virtual-memory normal multiple-selects

Answer key

1.28.2 Virtual Memory: GATE CSE 2004 | Question: 47



Consider a system with a two-level paging scheme in which a regular memory access takes 150 nanoseconds, and servicing a page fault takes 8 milliseconds. An average instruction takes 100 nanoseconds of CPU time, and two memory accesses. The TLB hit ratio is 90%, and the page fault rate is one in every 10,000 instructions. What is the effective average instruction execution time?

- A. 645 nanoseconds
C. 1215 nanoseconds

- B. 1050 nanoseconds
D. 1230 nanoseconds

gatecse-2004 co-and-architecture virtual-memory normal

Answer key

1.28.3 Virtual Memory: GATE CSE 2008 | Question: 38



In an instruction execution pipeline, the earliest that the data TLB (Translation Lookaside Buffer) can be accessed is:

- A. before effective address calculation has started
B. during effective address calculation
C. after effective address calculation has completed
D. after data cache lookup has completed

gatecse-2008 co-and-architecture virtual-memory normal

Answer key

Answer Keys

1.1.1	C	1.1.2	N/A	1.1.3	N/A	1.1.4	N/A	1.1.5	B
1.1.6	D	1.1.7	A;B;C;D	1.1.8	C	1.1.9	A	1.1.10	B
1.1.11	C	1.1.12	B	1.1.13	C	1.1.14	C	1.1.15	D
1.1.16	D	1.1.17	B	1.1.18	D	1.1.19	D	1.2.1	180
1.2.2	11.83:11.87	1.2.3	4:4	1.3.1	16417	1.4.1	A;B;C;D	1.4.2	D
1.5.1	N/A	1.5.2	N/A	1.5.3	22	1.5.4	N/A	1.5.5	D
1.5.6	D	1.5.7	61.25	1.5.8	N/A	1.5.9	B	1.5.10	B
1.5.11	N/A	1.5.12	N/A	1.5.13	C	1.5.14	A	1.5.15	A
1.5.16	D	1.5.17	C	1.5.18	B	1.5.19	D	1.5.20	C
1.5.21	A	1.5.22	A	1.5.23	D	1.5.24	B	1.5.25	C
1.5.26	D	1.5.27	C	1.5.28	C	1.5.29	D	1.5.30	C
1.5.31	A	1.5.32	A	1.5.33	A	1.5.34	D	1.5.35	D
1.5.36	20	1.5.37	1.68	1.5.38	14	1.5.39	A	1.5.40	24
1.5.41	30	1.5.42	0.05	1.5.43	14	1.5.44	A	1.5.45	4.7 : 4.8
1.5.46	18	1.5.47	B	1.5.48	D	1.5.49	160	1.5.50	13.3:13.3;13.5:13.5
1.5.51	B	1.5.52	17 : 17	1.5.53	2 : 2	1.5.54	A	1.5.55	A;B;D
1.5.56	0.85	1.5.57	19	1.5.58	B;C	1.5.59	3	1.5.60	A
1.5.61	B;C	1.5.62	C	1.5.63	B	1.5.64	C	1.5.65	C
1.5.66	A	1.5.67	B	1.5.68	D	1.5.69	A	1.6.1	76
1.7.1	B	1.8.1	D	1.8.2	B	1.8.3	456	1.8.4	80000 : 80000
1.8.5	A	1.8.6	C	1.8.7	C	1.8.8	C	1.9.1	A
1.10.1	A	1.10.2	B	1.10.3	B	1.10.4	B	1.11.1	B
1.12.1	N/A	1.12.2	D	1.12.3	B	1.12.4	B	1.12.5	28
1.12.6	C	1.12.7	A;B;C	1.13.1	A;B;D	1.13.2	A	1.13.3	A
1.13.4	A;B	1.13.5	128	1.14.1	77.3 : 77.3	1.15.1	95 : 95	1.16.1	True

1.16.2	True	1.16.3	False	1.16.4	A	1.16.5	1.4 : 1.5	1.16.6	B
1.16.7	B	1.16.8	A	1.17.1	False	1.17.2	N/A	1.17.3	C
1.17.4	C	1.17.5	A	1.17.6	-16.0	1.17.7	3.0:3.0	1.18.1	N/A
1.18.2	256	1.18.3	True	1.18.4	16383	1.18.5	500	1.18.6	32
1.18.7	14	1.18.8	32	1.18.9	34	1.18.10	D	1.18.11	B
1.19.1	D	1.20.1	D	1.20.2	B	1.20.3	A	1.20.4	A
1.20.5	B	1.20.6	C	1.20.7	C	1.20.8	10.2	1.20.9	A
1.20.10	C	1.21.1	N/A	1.21.2	N/A	1.21.3	N/A	1.21.4	B
1.21.5	A	1.21.6	D	1.21.7	B	1.21.8	C	1.21.9	B
1.21.10	D	1.21.11	A	1.21.12	C	1.21.13	D	1.21.14	D
1.21.15	16	1.21.16	50 : 50	1.21.17	B	1.21.18	B	1.21.19	D
1.21.20	C	1.21.21	A	1.22.1	TBA	1.22.2	N/A	1.22.3	D
1.22.4	31	1.22.5	59 : 60	1.22.6	C	1.23.1	N/A	1.23.2	C
1.23.3	C	1.23.4	B	1.23.5	D	1.23.6	A	1.23.7	D
1.23.8	A	1.23.9	D	1.23.10	B	1.23.11	D	1.23.12	D
1.24.1	15	1.24.2	B	1.24.3	N/A	1.24.4	N/A	1.24.5	D
1.24.6	D	1.24.7	C	1.24.8	C	1.24.9	B	1.24.10	A
1.24.11	D	1.24.12	B	1.24.13	B	1.24.14	B	1.24.15	C
1.24.16	B	1.24.17	1.50 : 1.60	1.24.18	C	1.24.19	3.2	1.24.20	13
1.24.21	3	1.24.22	33.0 : 34.0	1.24.23	4	1.24.24	1.50 : 1.51	1.24.25	219
1.24.26	2.15:2.18	1.24.27	17160 : 17160	1.24.28	1.87 : 1.88	1.24.29	2040	1.24.30	A;B;C
1.24.31	A	1.24.32	3	1.24.33	260.20:261.20	1.24.34	C	1.24.35	B
1.24.36	C	1.24.37	B	1.24.38	B	1.24.39	C	1.25.1	X
1.25.2	A	1.26.1	4	1.26.2	1.6	1.26.3	3	1.26.4	2.30:2.40
1.26.5	A	1.26.6	B	1.27.1	1.42:1.45	1.28.1	A;B	1.28.2	D
1.28.3	C								